Tugas Praktikum Pemrograman Mobile
Pertemuan Ke - 6



Nama : **Michael Cornelius Bong**
NIM : **20240801004**
Jurusan : **Ilmu Komputer**
Prodi : **Teknik Informatika**

Universitas Esa Unggul

Praktikum Flutter

1. Tujuan

Praktikum 6 bertujuan agar kita mampu membuat tema terang dan gelap menggunakan ThemeData dan ThemeMode, menerapkan named routes untuk berpindah halaman sambil mengirim data, membuat navigasi tab bawah menggunakan NavigationBar dan IndexedStack, serta menerapkan animasi menggunakan AnimatedContainer, AnimatedSwitcher, dan AnimationController.

2. Alat dan Bahan

- Komputer Atau Laptop
- Flutter Package (Flutter - Install)
- Android Studio / IDE Dengan Support Flutter & Dart
- Emulator SDK Android / HP Android Dengan Debugging Mode

3. Implementasi Tema / Theme Aplikasi

3.1 Pembuatan Tema Light and Dark Beserta Warna :

```
import 'package:flutter/material.dart';

final themeMode = ValueNotifier<ThemeMode>(ThemeMode.light);
final seedColor = ValueNotifier<Color>(Colors.indigo);

const pilihanWarna = [
 Colors.indigo,
 Colors.teal,
 Colors.deepOrange,
 Colors.pink,
];

ThemeData buatTema(Color seed, Brightness brightness) {
 return ThemeData(
   useMaterial3: true,
   colorSchemeSeed: seed,
   brightness: brightness,
```

```
    appBarTheme: const AppBarTheme(centerTitle: true),
  );
}

void main() => runApp(const MyApp());

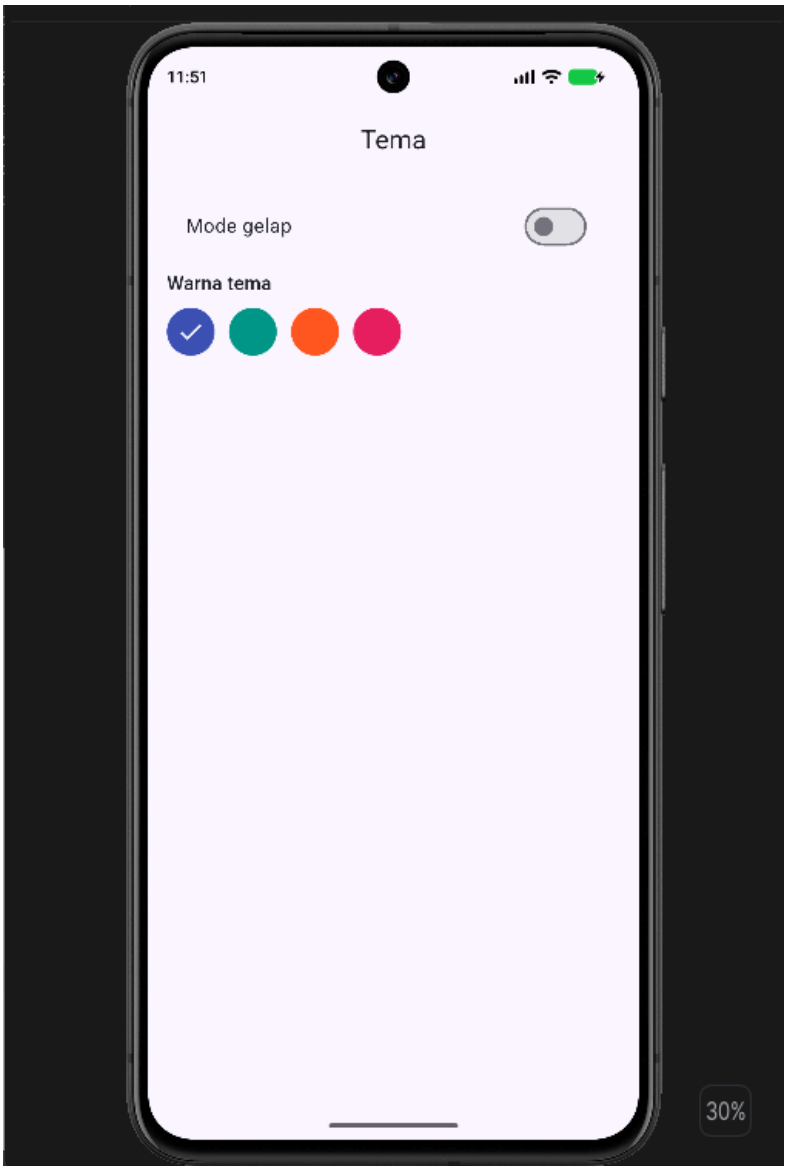
class MyApp extends StatelessWidget {
  const MyApp({super.key});

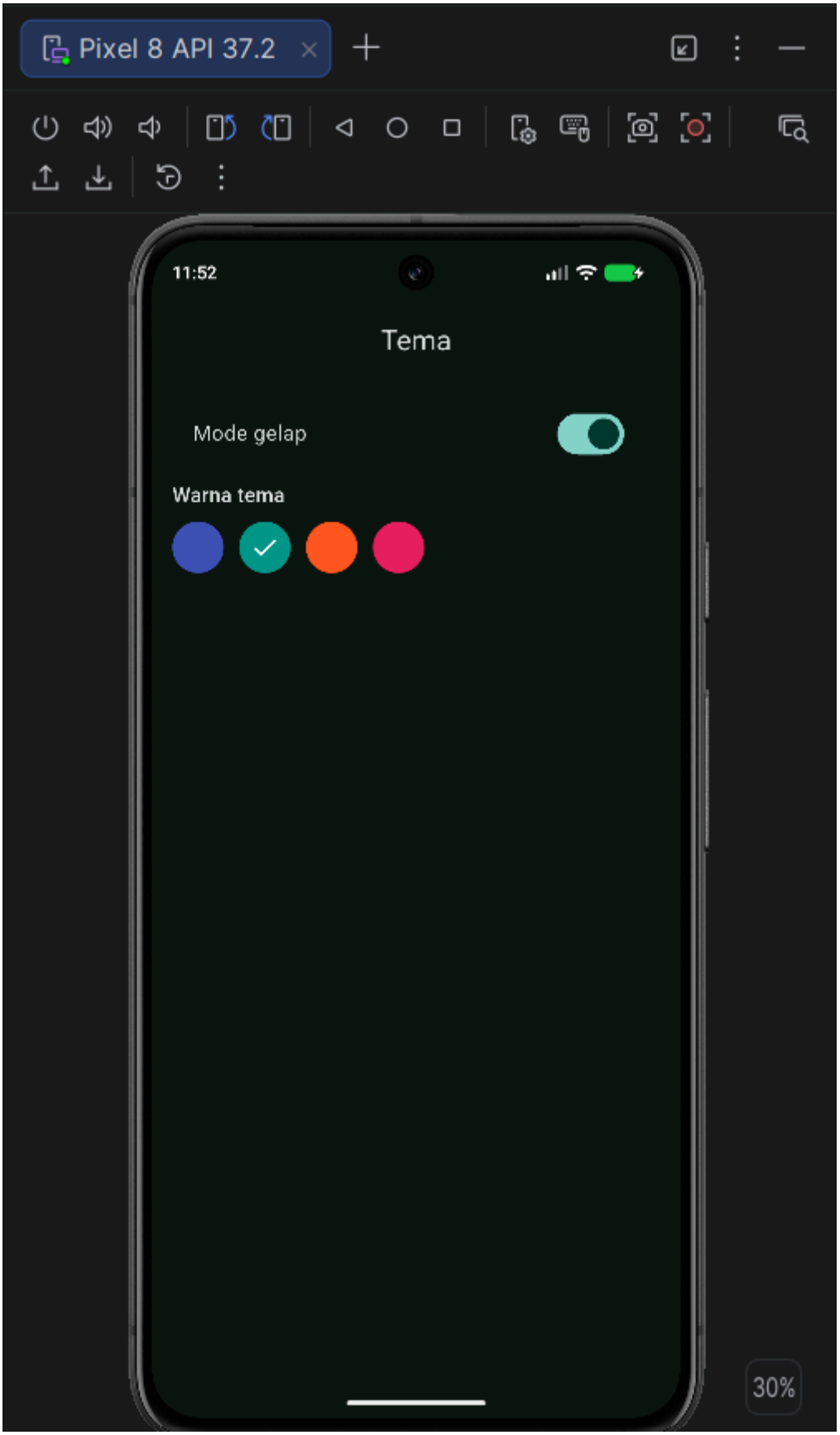
  @override
  Widget build(BuildContext context) {
    return ListenableBuilder(
      listenable: Listenable.merge([themeMode, seedColor]),
      builder: (context, _) {
        return MaterialApp(
          title: 'Praktikum 6',
          debugShowCheckedModeBanner: false,
          theme: buatTema(seedColor.value, Brightness.light),
          darkTheme: buatTema(seedColor.value, Brightness.dark),
          themeMode: themeMode.value,
          home: Scaffold(
            appBar: AppBar(title: const Text('Tema')),
            body: const PengaturanTab(),
          ),
        );
      },
    );
  }
}

class PengaturanTab extends StatelessWidget {
  const PengaturanTab({super.key});

  @override
  Widget build(BuildContext context) {
    return ListenableBuilder(
      listenable: Listenable.merge([themeMode, seedColor]),
      builder: (context, _) {
        return ListView(
          padding: const EdgeInsets.all(16),
          children: [
            SwitchListTile(
              title: const Text('Mode gelap'),
              value: themeMode.value == ThemeMode.dark,
              onChanged: (v) {
                themeMode.value = v ? ThemeMode.dark : ThemeMode.light;
              },
            ),
            const SizedBox(height: 8),
            Text(
              'Warna tema',
              style: Theme.of(context).textTheme.titleMedium,
            ),
```

```
            const SizedBox(height: 8),
            Wrap(
              spacing: 12,
              children: [
                for (final c in pilihanWarna)
                  GestureDetector(
                    onTap: () => seedColor.value = c,
                    child: CircleAvatar(
                      backgroundColor: c,
                      child: seedColor.value == c
                          ? const Icon(Icons.check, color: Colors.white)
                          : null,
                    ),
                  ),
              ],
            ),
          ],
        );
      },
    );
  }
}
```




Pixel 8 API 37.2
11:52
Tema
Mode gelap
Warna tema
30%

3.2 Implementasi Named Routes, Argumen, dan Hero :

```dart
import 'package:flutter/material.dart';

final themeMode = ValueNotifier<ThemeMode>(ThemeMode.light);
final seedColor = ValueNotifier<Color>(Colors.indigo);

const pilihanWarna = [
 Colors.indigo,
 Colors.teal,
 Colors.deepOrange,
 Colors.pink,
];

ThemeData buatTema(Color seed, Brightness brightness) {
 return ThemeData(
   useMaterial3: true,
   colorSchemeSeed: seed,
   brightness: brightness,
   appBarTheme: const AppBarTheme(centerTitle: true),
 );
}

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
 const MyApp({super.key});

 @override
 Widget build(BuildContext context) {
   return ListenableBuilder(
     listenable: Listenable.merge([themeMode, seedColor]),
     builder: (context, _) {
       return MaterialApp(
         title: 'Praktikum 6',
         debugShowCheckedModeBanner: false,
         theme: buatTema(seedColor.value, Brightness.light),
         darkTheme: buatTema(seedColor.value, Brightness.dark),
         themeMode: themeMode.value,
         initialRoute: '/',
         routes: {
           '/': (_) => Scaffold(
             appBar: AppBar(title: const Text('Beranda')),
             body: const BerandaTab(),
           ),
         },
         onGenerateRoute: (settings) {
           if (settings.name == '/detail') {
             final item = settings.arguments as Item;
             return MaterialPageRoute(
               builder: (_) => DetailPage(item: item),
               settings: settings,
```

```
            );
          }
          return null;
        },
      );
    },
  );
 }
}

class PengaturanTab extends StatelessWidget {
 const PengaturanTab({super.key});

 @override
 Widget build(BuildContext context) {
   return ListenableBuilder(
     listenable: Listenable.merge([themeMode, seedColor]),
     builder: (context, _) {
       return ListView(
         padding: const EdgeInsets.all(16),
         children: [
           SwitchListTile(
             title: const Text('Mode gelap'),
             value: themeMode.value == ThemeMode.dark,
             onChanged: (v) {
               themeMode.value = v ? ThemeMode.dark : ThemeMode.light;
             },
           ),
           const SizedBox(height: 8),
           Text('Warna tema', style:
Theme.of(context).textTheme.titleMedium),
           const SizedBox(height: 8),
           Wrap(
             spacing: 12,
             children: [
               for (final c in pilihanWarna)
                 GestureDetector(
                   onTap: () => seedColor.value = c,
                   child: CircleAvatar(
                     backgroundColor: c,
                     child: seedColor.value == c
                         ? const Icon(Icons.check, color: Colors.white)
                         : null,
                   ),
                 ),
             ],
           ),
         ],
       );
     },
   );
 }
}
```

```dart
class Item {
 final String nama;
 final IconData ikon;
 final Color warna;
 const Item(this.nama, this.ikon, this.warna);
}

const daftarItem = [
 Item('Flutter', Icons.flutter_dash, Colors.blue),
 Item('Musik', Icons.music_note, Colors.pink),
 Item('Kamera', Icons.camera_alt, Colors.orange),
 Item('Peta', Icons.map, Colors.green),
];

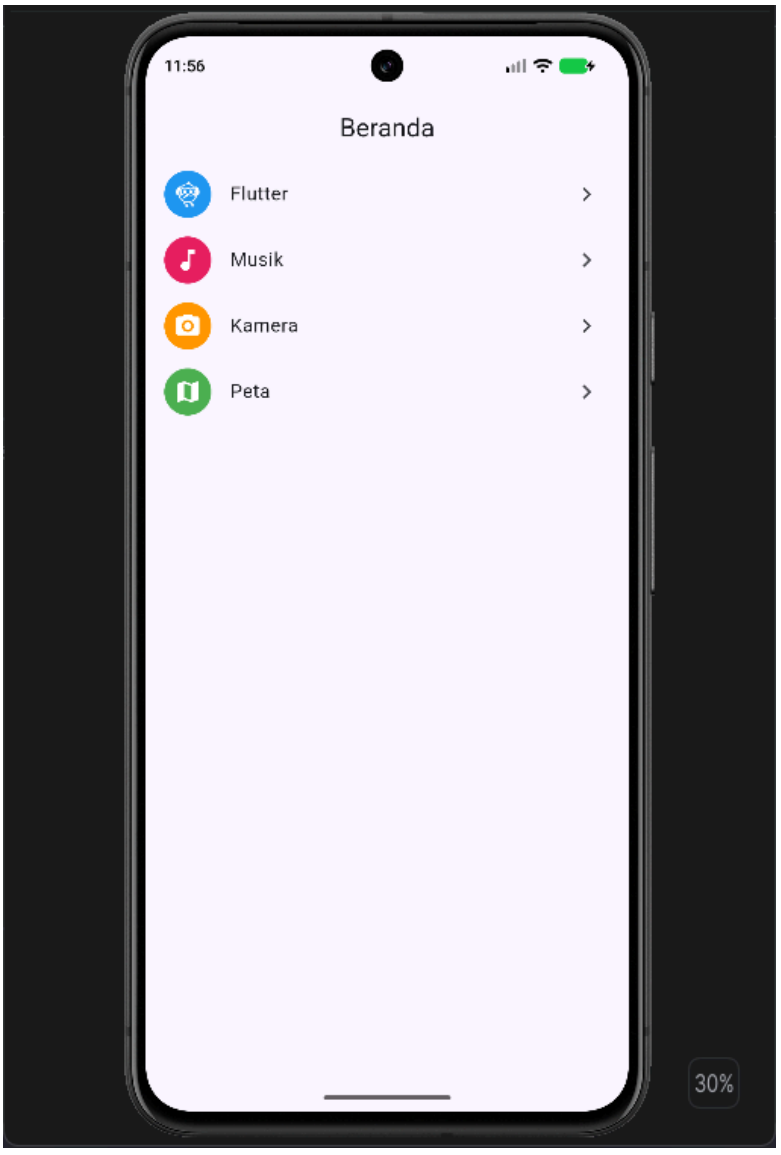
class BerandaTab extends StatelessWidget {
 const BerandaTab({super.key});

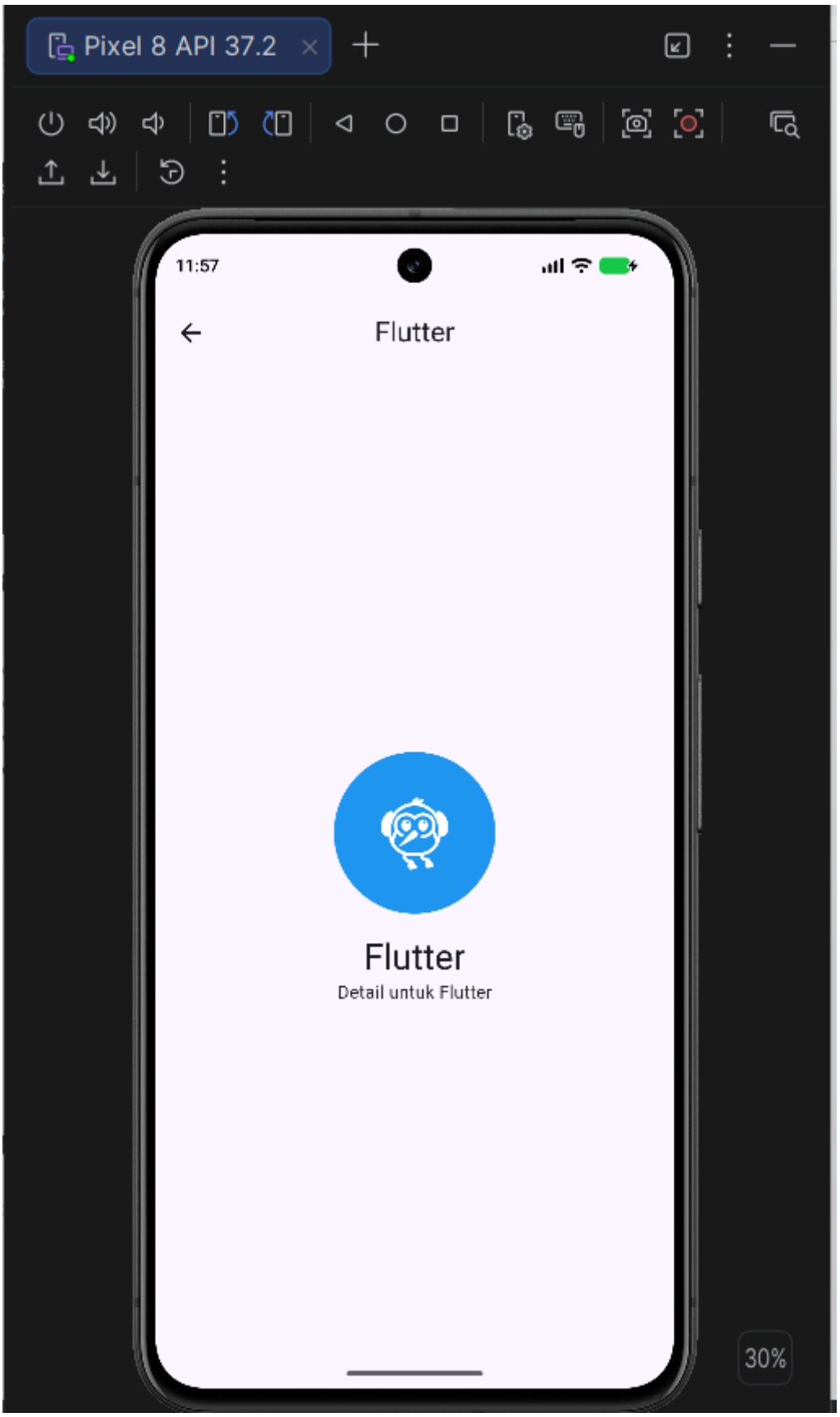
 @override
 Widget build(BuildContext context) {
   return ListView.builder(
     itemCount: daftarItem.length,
     itemBuilder: (context, i) {
       final item = daftarItem[i];
       return ListTile(
         leading: Hero(
           tag: 'ikon-${item.nama}',
           child: CircleAvatar(
             backgroundColor: item.warna,
             child: Icon(item.ikon, color: Colors.white),
           ),
         ),
         title: Text(item.nama),
         trailing: const Icon(Icons.chevron_right),
         onTap: () {
           Navigator.pushNamed(context, '/detail', arguments: item);
         },
       );
     },
   );
 }
}

class DetailPage extends StatelessWidget {
 final Item item;
 const DetailPage({super.key, required this.item});

 @override
 Widget build(BuildContext context) {
   final tema = Theme.of(context);
   return Scaffold(
     appBar: AppBar(title: Text(item.nama)),
     body: Center(
```

```
        child: Column(
          mainAxisSize: MainAxisSize.min,
          children: [
            Hero(
              tag: 'ikon-${item.nama}',
              child: CircleAvatar(
                radius: 64,
                backgroundColor: item.warna,
                child: Icon(item.ikon, size: 64, color: Colors.white),
              ),
            ),
            const SizedBox(height: 16),
            Text(item.nama, style: tema.textTheme.headlineMedium),
            Text('Detail untuk ${item.nama}', style:
tema.textTheme.bodyMedium),
          ],
        ),
      ),
    );
  }
}
```




Pixel 8 API 37.2
11:57
Flutter
Flutter
Detail untuk Flutter
30%

3.3 Penambahan Animasi :

```
import 'package:flutter/material.dart';

final themeMode = ValueNotifier<ThemeMode>(ThemeMode.light);
final seedColor = ValueNotifier<Color>(Colors.indigo);

const pilihanWarna = [
 Colors.indigo,
 Colors.teal,
 Colors.deepOrange,
 Colors.pink,
];

ThemeData buatTema(Color seed, Brightness brightness) {
 return ThemeData(
   useMaterial3: true,
   colorSchemeSeed: seed,
   brightness: brightness,
   appBarTheme: const AppBarTheme(centerTitle: true),
 );
}

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
 const MyApp({super.key});

 @override
 Widget build(BuildContext context) {
   return ListenableBuilder(
     listenable: Listenable.merge([themeMode, seedColor]),
     builder: (context, _) {
       return MaterialApp(
         title: 'Praktikum 6',
         debugShowCheckedModeBanner: false,
         theme: buatTema(seedColor.value, Brightness.light),
         darkTheme: buatTema(seedColor.value, Brightness.dark),
         themeMode: themeMode.value,
         initialRoute: '/',
         routes: {
           // Sementara diisi AnimasiTab untuk mencoba Bagian C,
           // nanti di Bagian D diganti ShellPage.
           '/': (_) => Scaffold(
             appBar: AppBar(title: const Text('Animasi')),
             body: const AnimasiTab(),
           ),
         },
         onGenerateRoute: (settings) {
           if (settings.name == '/detail') {
             final item = settings.arguments as Item;
             return MaterialPageRoute(
```

```
              builder: (_) => DetailPage(item: item),
              settings: settings,
            );
          }
          return null;
        },
      );
    },
  );
 }
}

class PengaturanTab extends StatelessWidget {
 const PengaturanTab({super.key});

 @override
 Widget build(BuildContext context) {
   return ListenableBuilder(
     listenable: Listenable.merge([themeMode, seedColor]),
     builder: (context, _) {
       return ListView(
         padding: const EdgeInsets.all(16),
         children: [
           SwitchListTile(
             title: const Text('Mode gelap'),
             value: themeMode.value == ThemeMode.dark,
             onChanged: (v) {
               themeMode.value = v ? ThemeMode.dark : ThemeMode.light;
             },
           ),
           const SizedBox(height: 8),
           Text('Warna tema', style:
Theme.of(context).textTheme.titleMedium),
           const SizedBox(height: 8),
           Wrap(
             spacing: 12,
             children: [
               for (final c in pilihanWarna)
                 GestureDetector(
                   onTap: () => seedColor.value = c,
                   child: CircleAvatar(
                     backgroundColor: c,
                     child: seedColor.value == c
                         ? const Icon(Icons.check, color: Colors.white)
                         : null,
                   ),
                 ),
             ],
           ),
         ],
       );
     },
   );
```

```
 }
}

class Item {
 final String nama;
 final IconData ikon;
 final Color warna;
 const Item(this.nama, this.ikon, this.warna);
}

const daftarItem = [
 Item('Flutter', Icons.flutter_dash, Colors.blue),
 Item('Musik', Icons.music_note, Colors.pink),
 Item('Kamera', Icons.camera_alt, Colors.orange),
 Item('Peta', Icons.map, Colors.green),
];

class BerandaTab extends StatelessWidget {
 const BerandaTab({super.key});

 @override
 Widget build(BuildContext context) {
   return ListView.builder(
     itemCount: daftarItem.length,
     itemBuilder: (context, i) {
       final item = daftarItem[i];
       return ListTile(
         leading: Hero(
           tag: 'ikon-${item.nama}',
           child: CircleAvatar(
             backgroundColor: item.warna,
             child: Icon(item.ikon, color: Colors.white),
           ),
         ),
         title: Text(item.nama),
         trailing: const Icon(Icons.chevron_right),
         onTap: () {
           Navigator.pushNamed(context, '/detail', arguments: item);
         },
       );
     },
   );
 }
}

class DetailPage extends StatelessWidget {
 final Item item;
 const DetailPage({super.key, required this.item});

 @override
 Widget build(BuildContext context) {
   final tema = Theme.of(context);
   return Scaffold(
```

```
    appBar: AppBar(title: Text(item.nama)),
    body: Center(
      child: Column(
        mainAxisSize: MainAxisSize.min,
        children: [
          Hero(
            tag: 'ikon-${item.nama}',
            child: CircleAvatar(
              radius: 64,
              backgroundColor: item.warna,
              child: Icon(item.ikon, size: 64, color: Colors.white),
            ),
          ),
          const SizedBox(height: 16),
          Text(item.nama, style: tema.textTheme.headlineMedium),
          Text('Detail untuk ${item.nama}', style:
tema.textTheme.bodyMedium),
        ],
      ),
    ),
  );
 }
}

class AnimasiTab extends StatefulWidget {
 const AnimasiTab({super.key});

 @override
 State<AnimasiTab> createState() => _AnimasiTabState();
}

class _AnimasiTabState extends State<AnimasiTab>
   with SingleTickerProviderStateMixin {
 bool _besar = false;
 int _hitung = 0;
 late final AnimationController _putar;

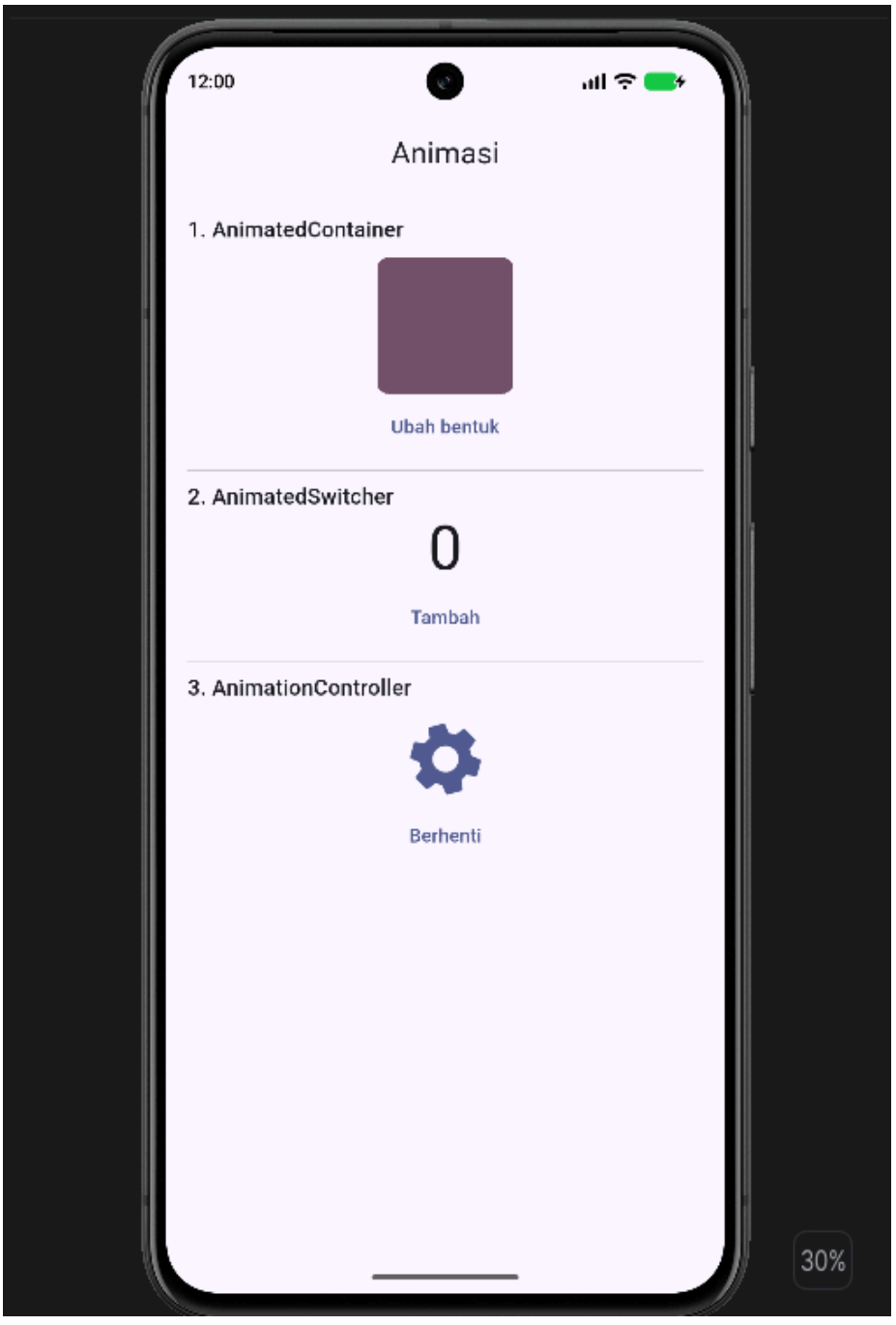
 @override
 void initState() {
   super.initState();
   _putar = AnimationController(
     vsync: this,
     duration: const Duration(seconds: 2),
   )..repeat();
 }

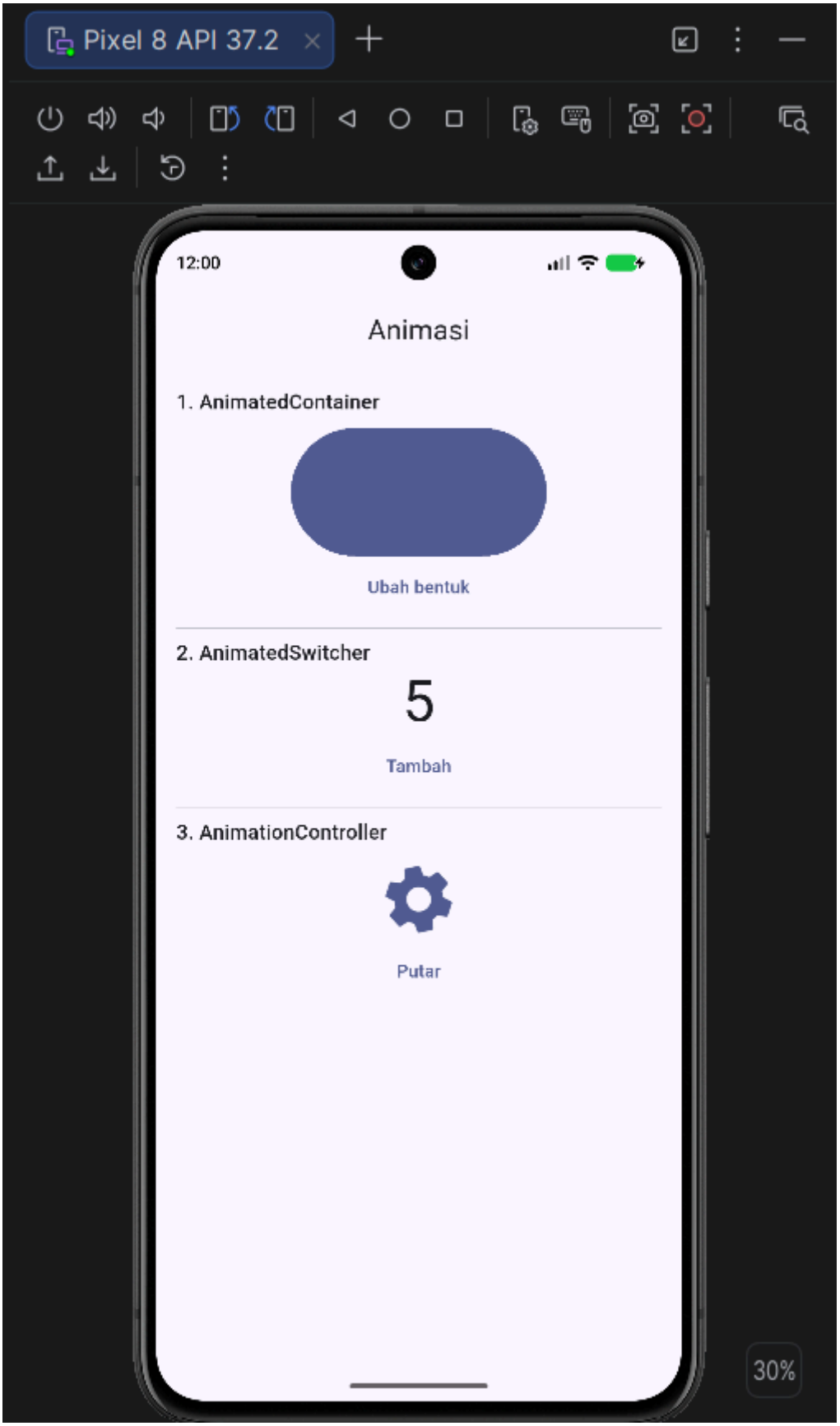
 @override
 void dispose() {
   _putar.dispose();
   super.dispose();
 }

 @override
```

```dart
Widget build(BuildContext context) {
  final tema = Theme.of(context);
  final warna = tema.colorScheme;
  return ListView(
    padding: const EdgeInsets.all(16),
    children: [
      Text('1. AnimatedContainer', style: tema.textTheme.titleMedium),
      const SizedBox(height: 8),
      Center(
        child: AnimatedContainer(
          duration: const Duration(milliseconds: 500),
          curve: Curves.easeInOut,
          width: _besar ? 200 : 100,
          height: 100,
          decoration: BoxDecoration(
            color: _besar ? warna.primary : warna.tertiary,
            borderRadius: BorderRadius.circular(_besar ? 50 : 8),
          ),
        ),
      ),
      TextButton(
        onPressed: () => setState(() => _besar = !_besar),
        child: const Text('Ubah bentuk'),
      ),
      const Divider(),
      Text('2. AnimatedSwitcher', style: tema.textTheme.titleMedium),
      Center(
        child: AnimatedSwitcher(
          duration: const Duration(milliseconds: 300),
          transitionBuilder: (child, anim) {
            return ScaleTransition(scale: anim, child: child);
          },
          child: Text(
            '$_hitung',
            key: ValueKey(_hitung),
            style: tema.textTheme.displayMedium,
          ),
        ),
      ),
      TextButton(
        onPressed: () => setState(() => _hitung++),
        child: const Text('Tambah'),
      ),
      const Divider(),
      Text('3. AnimationController', style: tema.textTheme.titleMedium),
      const SizedBox(height: 8),
      Center(
        child: RotationTransition(
          turns: _putar,
          child: Icon(Icons.settings, size: 64, color: warna.primary),
        ),
      ),
      TextButton(
```

```
          onPressed: () {
            if (_putar.isAnimating) {
              _putar.stop();
            } else {
              _putar.repeat();
            }
            setState(() {});
          },
          child: Text(_putar.isAnimating ? 'Berhenti' : 'Putar'),
        ),
      ],
    );
  }
}
```




Pixel 8 API 37.2
12:00
Animasi
1. AnimatedContainer
Ubah bentuk
2. AnimatedSwitcher
5
Tambah
3. AnimationController
Putar
30%

3.4 Navigasi Tab Bawah Yang Menyatukan Semua :

```
import 'package:flutter/material.dart';

final themeMode = ValueNotifier<ThemeMode>(ThemeMode.light);
final seedColor = ValueNotifier<Color>(Colors.indigo);

const pilihanWarna = [
 Colors.indigo,
 Colors.teal,
 Colors.deepOrange,
 Colors.pink,
];

ThemeData buatTema(Color seed, Brightness brightness) {
 return ThemeData(
   useMaterial3: true,
   colorSchemeSeed: seed,
   brightness: brightness,
   appBarTheme: const AppBarTheme(centerTitle: true),
 );
}

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
 const MyApp({super.key});

 @override
 Widget build(BuildContext context) {
   return ListenableBuilder(
     listenable: Listenable.merge([themeMode, seedColor]),
     builder: (context, _) {
       return MaterialApp(
         title: 'Praktikum 6',
         debugShowCheckedModeBanner: false,
         theme: buatTema(seedColor.value, Brightness.light),
         darkTheme: buatTema(seedColor.value, Brightness.dark),
         themeMode: themeMode.value,
         initialRoute: '/',
         routes: {
           '/': (_) => const ShellPage(),
         },
         onGenerateRoute: (settings) {
           if (settings.name == '/detail') {
             final item = settings.arguments as Item;
             return MaterialPageRoute(
               builder: (_) => DetailPage(item: item),
               settings: settings,
             );
           }
           return null;
```

```
        },
      );
    },
  );
 }
}

class PengaturanTab extends StatelessWidget {
 const PengaturanTab({super.key});

 @override
 Widget build(BuildContext context) {
   return ListenableBuilder(
     listenable: Listenable.merge([themeMode, seedColor]),
     builder: (context, _) {
       return ListView(
         padding: const EdgeInsets.all(16),
         children: [
           SwitchListTile(
             title: const Text('Mode gelap'),
             value: themeMode.value == ThemeMode.dark,
             onChanged: (v) {
               themeMode.value = v ? ThemeMode.dark : ThemeMode.light;
             },
           ),
           const SizedBox(height: 8),
           Text('Warna tema', style:
Theme.of(context).textTheme.titleMedium),
           const SizedBox(height: 8),
           Wrap(
             spacing: 12,
             children: [
               for (final c in pilihanWarna)
                 GestureDetector(
                   onTap: () => seedColor.value = c,
                   child: CircleAvatar(
                     backgroundColor: c,
                     child: seedColor.value == c
                         ? const Icon(Icons.check, color: Colors.white)
                         : null,
                   ),
                 ),
             ],
           ),
         ],
       );
     },
   );
 }
}

class Item {
 final String nama;
```

```dart
 final IconData ikon;
 final Color warna;
 const Item(this.nama, this.ikon, this.warna);
}

const daftarItem = [
 Item('Flutter', Icons.flutter_dash, Colors.blue),
 Item('Musik', Icons.music_note, Colors.pink),
 Item('Kamera', Icons.camera_alt, Colors.orange),
 Item('Peta', Icons.map, Colors.green),
];

class BerandaTab extends StatelessWidget {
 const BerandaTab({super.key});

 @override
 Widget build(BuildContext context) {
   return ListView.builder(
     itemCount: daftarItem.length,
     itemBuilder: (context, i) {
       final item = daftarItem[i];
       return ListTile(
         leading: Hero(
           tag: 'ikon-${item.nama}',
           child: CircleAvatar(
             backgroundColor: item.warna,
             child: Icon(item.ikon, color: Colors.white),
           ),
         ),
         title: Text(item.nama),
         trailing: const Icon(Icons.chevron_right),
         onTap: () {
           Navigator.pushNamed(context, '/detail', arguments: item);
         },
       );
     },
   );
 }
}

class DetailPage extends StatelessWidget {
 final Item item;
 const DetailPage({super.key, required this.item});

 @override
 Widget build(BuildContext context) {
   final tema = Theme.of(context);
   return Scaffold(
     appBar: AppBar(title: Text(item.nama)),
     body: Center(
       child: Column(
         mainAxisSize: MainAxisSize.min,
         children: [
```

```
          Hero(
            tag: 'ikon-${item.nama}',
            child: CircleAvatar(
              radius: 64,
              backgroundColor: item.warna,
              child: Icon(item.ikon, size: 64, color: Colors.white),
            ),
          ),
          const SizedBox(height: 16),
          Text(item.nama, style: tema.textTheme.headlineMedium),
          Text('Detail untuk ${item.nama}', style:
tema.textTheme.bodyMedium),
        ],
      ),
    ),
  );
 }
}

class AnimasiTab extends StatefulWidget {
 const AnimasiTab({super.key});

 @override
 State<AnimasiTab> createState() => _AnimasiTabState();
}

class _AnimasiTabState extends State<AnimasiTab>
   with SingleTickerProviderStateMixin {
 bool _besar = false;
 int _hitung = 0;
 late final AnimationController _putar;

 @override
 void initState() {
   super.initState();
   _putar = AnimationController(
     vsync: this,
     duration: const Duration(seconds: 2),
   )..repeat();
 }

 @override
 void dispose() {
   _putar.dispose();
   super.dispose();
 }

 @override
 Widget build(BuildContext context) {
   final tema = Theme.of(context);
   final warna = tema.colorScheme;
   return ListView(
     padding: const EdgeInsets.all(16),
```

```
    children: [
      Text('1. AnimatedContainer', style: tema.textTheme.titleMedium),
      const SizedBox(height: 8),
      Center(
        child: AnimatedContainer(
          duration: const Duration(milliseconds: 500),
          curve: Curves.easeInOut,
          width: _besar ? 200 : 100,
          height: 100,
          decoration: BoxDecoration(
            color: _besar ? warna.primary : warna.tertiary,
            borderRadius: BorderRadius.circular(_besar ? 50 : 8),
          ),
        ),
      ),
      TextButton(
        onPressed: () => setState(() => _besar = !_besar),
        child: const Text('Ubah bentuk'),
      ),
      const Divider(),
      Text('2. AnimatedSwitcher', style: tema.textTheme.titleMedium),
      Center(
        child: AnimatedSwitcher(
          duration: const Duration(milliseconds: 300),
          transitionBuilder: (child, anim) {
            return ScaleTransition(scale: anim, child: child);
          },
          child: Text(
            '$_hitung',
            key: ValueKey(_hitung),
            style: tema.textTheme.displayMedium,
          ),
        ),
      ),
      TextButton(
        onPressed: () => setState(() => _hitung++),
        child: const Text('Tambah'),
      ),
      const Divider(),
      Text('3. AnimationController', style: tema.textTheme.titleMedium),
      const SizedBox(height: 8),
      Center(
        child: RotationTransition(
          turns: _putar,
          child: Icon(Icons.settings, size: 64, color: warna.primary),
        ),
      ),
      TextButton(
        onPressed: () {
          if (_putar.isAnimating) {
            _putar.stop();
          } else {
            _putar.repeat();
```

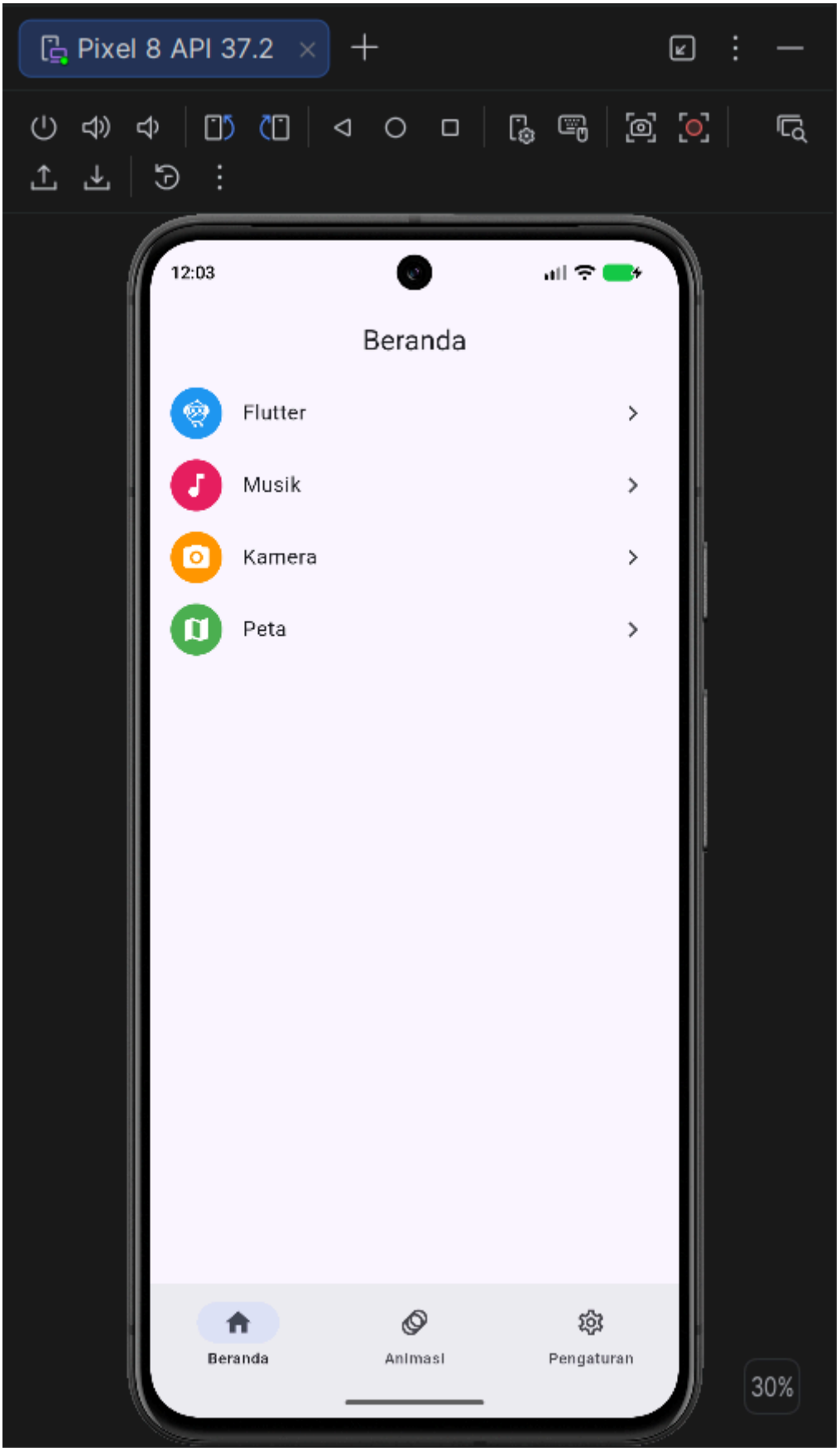
```
            }
            setState(() {});
          },
          child: Text(_putar.isAnimating ? 'Berhenti' : 'Putar'),
        ),
      ],
    );
  }
}

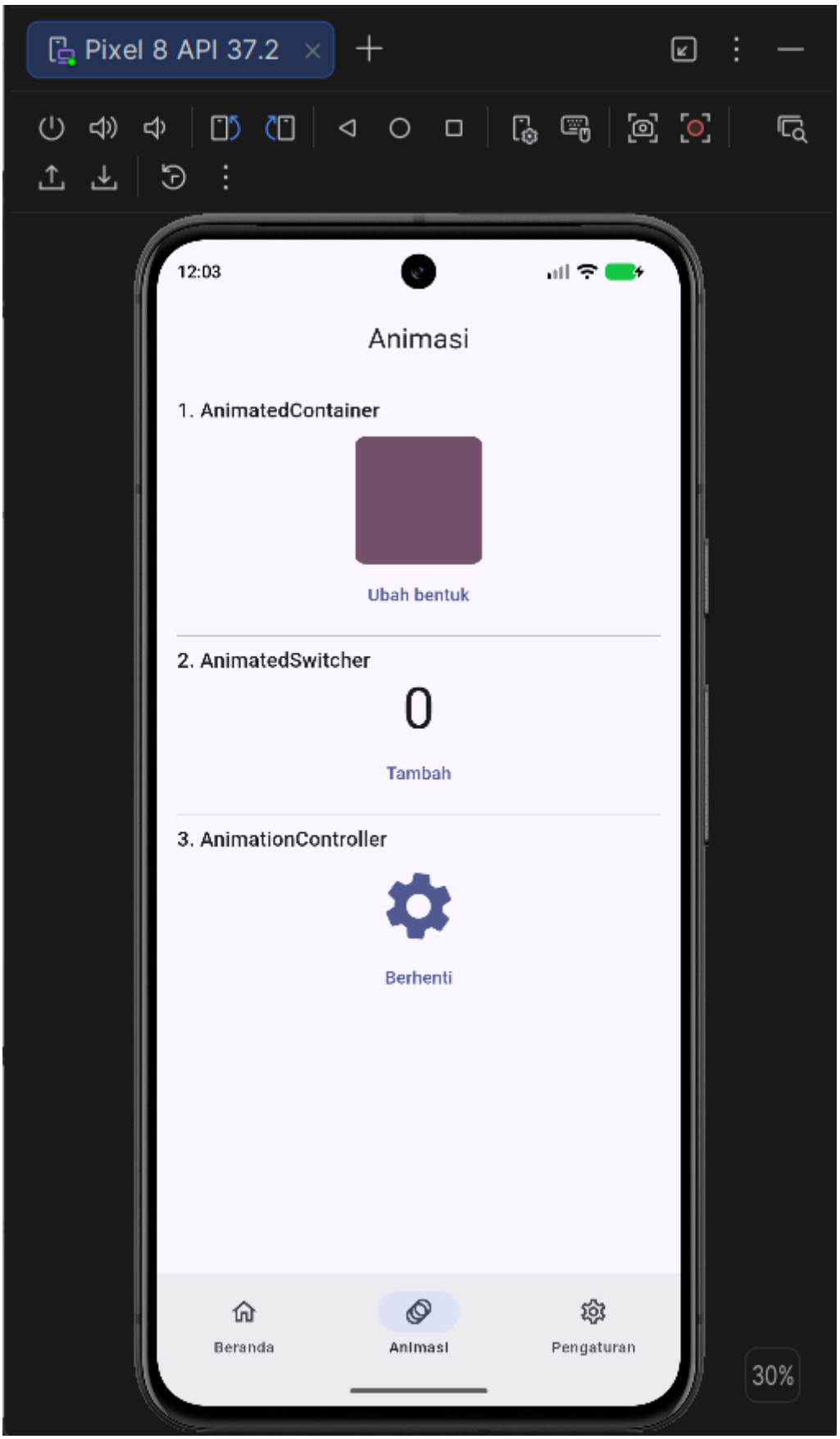
class ShellPage extends StatefulWidget {
  const ShellPage({super.key});

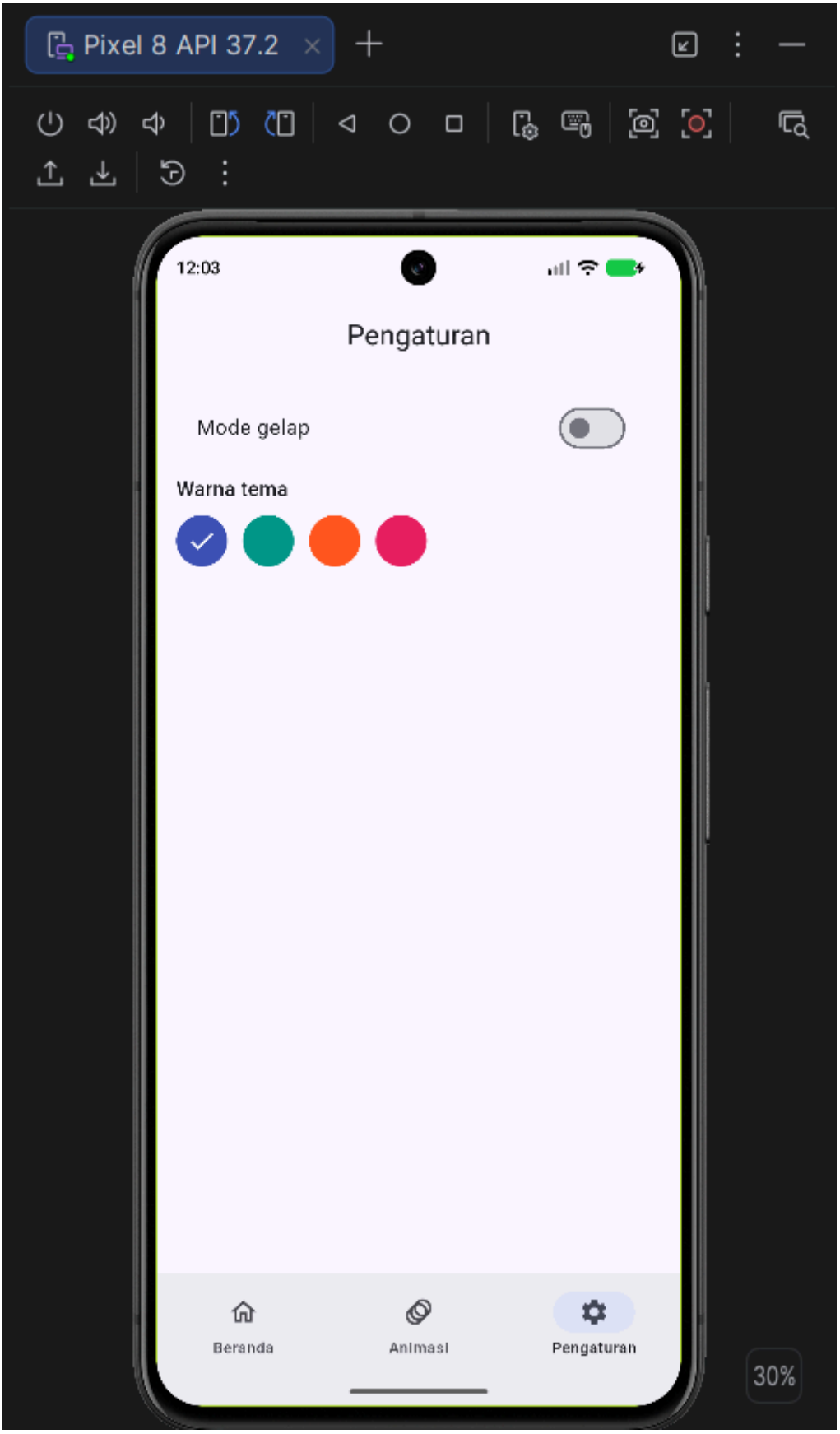
  @override
  State<ShellPage> createState() => _ShellPageState();
}

class _ShellPageState extends State<ShellPage> {
  int _index = 0;
  static const _halaman = [BerandaTab(), AnimasiTab(), PengaturanTab()];
  static const _judul = ['Beranda', 'Animasi', 'Pengaturan'];

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text(_judul[_index])),
      body: IndexedStack(index: _index, children: _halaman),
      bottomNavigationBar: NavigationBar(
        selectedIndex: _index,
        onDestinationSelected: (i) => setState(() => _index = i),
        destinations: const [
          NavigationDestination(
            icon: Icon(Icons.home_outlined),
            selectedIcon: Icon(Icons.home),
            label: 'Beranda',
          ),
          NavigationDestination(
            icon: Icon(Icons.animation),
            label: 'Animasi',
          ),
          NavigationDestination(
            icon: Icon(Icons.settings_outlined),
            selectedIcon: Icon(Icons.settings),
            label: 'Pengaturan',
          ),
        ],
      ),
    );
  }
}
```


Pixel 8 API 37.2
12:03
Beranda
Flutter
Musik
Kamera
Peta
Beranda
Animasi
Pengaturan
30%


Pixel 8 API 37.2
12:03
Animasi
1. AnimatedContainer
Ubah bentuk
2. AnimatedSwitcher
0
Tambah
3. AnimationController
Berhenti
Beranda
Animasi
Pengaturan
30%


Pixel 8 API 37.2
12:03
Pengaturan
Mode gelap
Warna tema
Beranda
Animasi
Pengaturan
30%

4. Latihan Mandiri

1. Simpan mode gelap dan warna tema dengan shared_preferences (Pertemuan 5) dan muat saat aplikasi dimulai (WidgetsFlutterBinding.ensureInitialized() di main()).
2. Tambahkan Drawer pada ShellPage berisi menu untuk berpindah tab.
3. Ganti transisi halaman detail dengan PageRouteBuilder yang memakai FadeTransition (di dalam onGenerateRoute).
4. Tambahkan onUnknownRoute yang menampilkan halaman "404 - Halaman tidak ditemukan".
5. Tambahkan demo keempat di AnimasiTab: AnimatedOpacity yang memudarkan dan memunculkan sebuah kartu saat tombol ditekan.

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

final themeMode = ValueNotifier<ThemeMode>(ThemeMode.light);
final seedColor = ValueNotifier<Color>(Colors.indigo);

const pilihanWarna = [
 Colors.indigo,
 Colors.teal,
 Colors.deepOrange,
 Colors.pink,
];

ThemeData buatTema(Color seed, Brightness brightness) {
 return ThemeData(
   useMaterial3: true,
   colorSchemeSeed: seed,
   brightness: brightness,
   appBarTheme: const AppBarTheme(centerTitle: true),
 );
}

Future<void> main() async {
 WidgetsFlutterBinding.ensureInitialized();
 final prefs = await SharedPreferences.getInstance();
 final modeGelap = prefs.getBool('modeGelap') ?? false;
 final indeksWarna =
     (prefs.getInt('warnaTema') ?? 0).clamp(0, pilihanWarna.length -
1).toInt();

 themeMode.value = modeGelap ? ThemeMode.dark : ThemeMode.light;
 seedColor.value = pilihanWarna[indeksWarna];

 runApp(const MyApp());
}

class MyApp extends StatelessWidget {
 const MyApp({super.key});
```

```
 @override
 Widget build(BuildContext context) {
   return ListenableBuilder(
     listenable: Listenable.merge([themeMode, seedColor]),
     builder: (context, _) {
       return MaterialApp(
         title: 'Praktikum 6',
         debugShowCheckedModeBanner: false,
         theme: buatTema(seedColor.value, Brightness.light),
         darkTheme: buatTema(seedColor.value, Brightness.dark),
         themeMode: themeMode.value,
         initialRoute: '/',
         routes: {
           '/': (_) => const ShellPage(),
         },
         onGenerateRoute: (settings) {
           if (settings.name == '/detail') {
             final item = settings.arguments as Item;
             return PageRouteBuilder<void>(
               settings: settings,
               pageBuilder: (context, animation, secondaryAnimation) {
                 return DetailPage(item: item);
               },
               transitionsBuilder:
                   (context, animation, secondaryAnimation, child) {
                 return FadeTransition(
                   opacity: animation,
                   child: child,
                 );
               },
               transitionDuration: const Duration(milliseconds: 300),
             );
           }
           return null;
         },
         onUnknownRoute: (settings) {
           return MaterialPageRoute<void>(
             builder: (_) => const NotFoundPage(),
           );
         },
       );
     },
   );
 }
}

class PengaturanTab extends StatelessWidget {
 const PengaturanTab({super.key});

 @override
 Widget build(BuildContext context) {
   return ListenableBuilder(
```

```
    listenable: Listenable.merge([themeMode, seedColor]),
    builder: (context, _) {
      return ListView(
        padding: const EdgeInsets.all(16),
        children: [
          SwitchListTile(
            title: const Text('Mode gelap'),
            value: themeMode.value == ThemeMode.dark,
            onChanged: (v) async {
              themeMode.value = v ? ThemeMode.dark : ThemeMode.light;
              final prefs = await SharedPreferences.getInstance();
              await prefs.setBool('modeGelap', v);
            },
          ),
          const SizedBox(height: 8),
          Text(
            'Warna tema',
            style: Theme.of(context).textTheme.titleMedium,
          ),
          const SizedBox(height: 8),
          Wrap(
            spacing: 12,
            children: [
              for (final c in pilihanWarna)
                GestureDetector(
                  onTap: () async {
                    seedColor.value = c;
                    final prefs = await SharedPreferences.getInstance();
                    await prefs.setInt('warnaTema',
pilihanWarna.indexOf(c));
                  },
                  child: CircleAvatar(
                    backgroundColor: c,
                    child: seedColor.value == c
                        ? const Icon(Icons.check, color: Colors.white)
                        : null,
                  ),
                ),
            ],
          ),
        ],
      );
    },
  );
 }
}

class Item {
 final String nama;
 final IconData ikon;
 final Color warna;

 const Item(this.nama, this.ikon, this.warna);
```

```dart
}

const daftarItem = [
 Item('Flutter', Icons.flutter_dash, Colors.blue),
 Item('Musik', Icons.music_note, Colors.pink),
 Item('Kamera', Icons.camera_alt, Colors.orange),
 Item('Peta', Icons.map, Colors.green),
];

class BerandaTab extends StatelessWidget {
 const BerandaTab({super.key});

 @override
 Widget build(BuildContext context) {
   return ListView.builder(
     itemCount: daftarItem.length,
     itemBuilder: (context, i) {
       final item = daftarItem[i];
       return ListTile(
         leading: Hero(
           tag: 'ikon-${item.nama}',
           child: CircleAvatar(
             backgroundColor: item.warna,
             child: Icon(item.ikon, color: Colors.white),
           ),
         ),
         title: Text(item.nama),
         trailing: const Icon(Icons.chevron_right),
         onTap: () {
           Navigator.pushNamed(context, '/detail', arguments: item);
         },
       );
     },
   );
 }
}

class DetailPage extends StatelessWidget {
 final Item item;

 const DetailPage({super.key, required this.item});

 @override
 Widget build(BuildContext context) {
   final tema = Theme.of(context);
   return Scaffold(
     appBar: AppBar(title: Text(item.nama)),
     body: Center(
       child: Column(
         mainAxisSize: MainAxisSize.min,
         children: [
           Hero(
             tag: 'ikon-${item.nama}',
```

```
            child: CircleAvatar(
              radius: 64,
              backgroundColor: item.warna,
              child: Icon(item.ikon, size: 64, color: Colors.white),
            ),
          ),
          const SizedBox(height: 16),
          Text(item.nama, style: tema.textTheme.headlineMedium),
          Text('Detail untuk ${item.nama}', style:
tema.textTheme.bodyMedium),
        ],
      ),
    ),
  );
 }
}

class AnimasiTab extends StatefulWidget {
 const AnimasiTab({super.key});

 @override
 State<AnimasiTab> createState() => _AnimasiTabState();
}

class _AnimasiTabState extends State<AnimasiTab>
   with SingleTickerProviderStateMixin {
 bool _besar = false;
 bool _terlihat = true;
 int _hitung = 0;
 late final AnimationController _putar;

 @override
 void initState() {
   super.initState();
   _putar = AnimationController(
     vsync: this,
     duration: const Duration(seconds: 2),
   )..repeat();
 }

 @override
 void dispose() {
   _putar.dispose();
   super.dispose();
 }

 @override
 Widget build(BuildContext context) {
   final tema = Theme.of(context);
   final warna = tema.colorScheme;

   return ListView(
     padding: const EdgeInsets.all(16),
```

```
    children: [
      Text('1. AnimatedContainer', style: tema.textTheme.titleMedium),
      const SizedBox(height: 8),
      Center(
        child: AnimatedContainer(
          duration: const Duration(milliseconds: 500),
          curve: Curves.easeInOut,
          width: _besar ? 200 : 100,
          height: 100,
          decoration: BoxDecoration(
            color: _besar ? warna.primary : warna.tertiary,
            borderRadius: BorderRadius.circular(_besar ? 50 : 8),
          ),
        ),
      ),
      TextButton(
        onPressed: () => setState(() => _besar = !_besar),
        child: const Text('Ubah bentuk'),
      ),
      const Divider(),
      Text('2. AnimatedSwitcher', style: tema.textTheme.titleMedium),
      Center(
        child: AnimatedSwitcher(
          duration: const Duration(milliseconds: 300),
          transitionBuilder: (child, anim) {
            return ScaleTransition(scale: anim, child: child);
          },
          child: Text(
            '$_hitung',
            key: ValueKey(_hitung),
            style: tema.textTheme.displayMedium,
          ),
        ),
      ),
      TextButton(
        onPressed: () => setState(() => _hitung++),
        child: const Text('Tambah'),
      ),
      const Divider(),
      Text('3. AnimationController', style: tema.textTheme.titleMedium),
      const SizedBox(height: 8),
      Center(
        child: RotationTransition(
          turns: _putar,
          child: Icon(Icons.settings, size: 64, color: warna.primary),
        ),
      ),
      TextButton(
        onPressed: () {
          if (_putar.isAnimating) {
            _putar.stop();
          } else {
            _putar.repeat();
```

```
          }
          setState(() {});
        },
        child: Text(_putar.isAnimating ? 'Berhenti' : 'Putar'),
      ),
      const Divider(),
      Text('4. AnimatedOpacity', style: tema.textTheme.titleMedium),
      const SizedBox(height: 8),
      Center(
        child: AnimatedOpacity(
          opacity: _terlihat ? 1.0 : 0.0,
          duration: const Duration(milliseconds: 400),
          child: Card(
            color: warna.primaryContainer,
            child: Padding(
              padding: const EdgeInsets.all(20),
              child: Row(
                mainAxisSize: MainAxisSize.min,
                children: [
                  Icon(Icons.star, color: warna.onPrimaryContainer),
                  const SizedBox(width: 8),
                  Text(
                    'Kartu animasi',
                    style: tema.textTheme.titleMedium?.copyWith(
                      color: warna.onPrimaryContainer,
                    ),
                  ),
                ],
              ),
            ),
          ),
        ),
      ),
      TextButton(
        onPressed: () => setState(() => _terlihat = !_terlihat),
        child: Text(_terlihat ? 'Sembunyikan kartu' : 'Tampilkan kartu'),
      ),
    ],
  );
 }
}

class ShellPage extends StatefulWidget {
 const ShellPage({super.key});

 @override
 State<ShellPage> createState() => _ShellPageState();
}

class _ShellPageState extends State<ShellPage> {
 int _index = 0;

 static const _halaman = [
```

```
  BerandaTab(),
  AnimasiTab(),
  PengaturanTab(),
];

static const _judul = ['Beranda', 'Animasi', 'Pengaturan'];

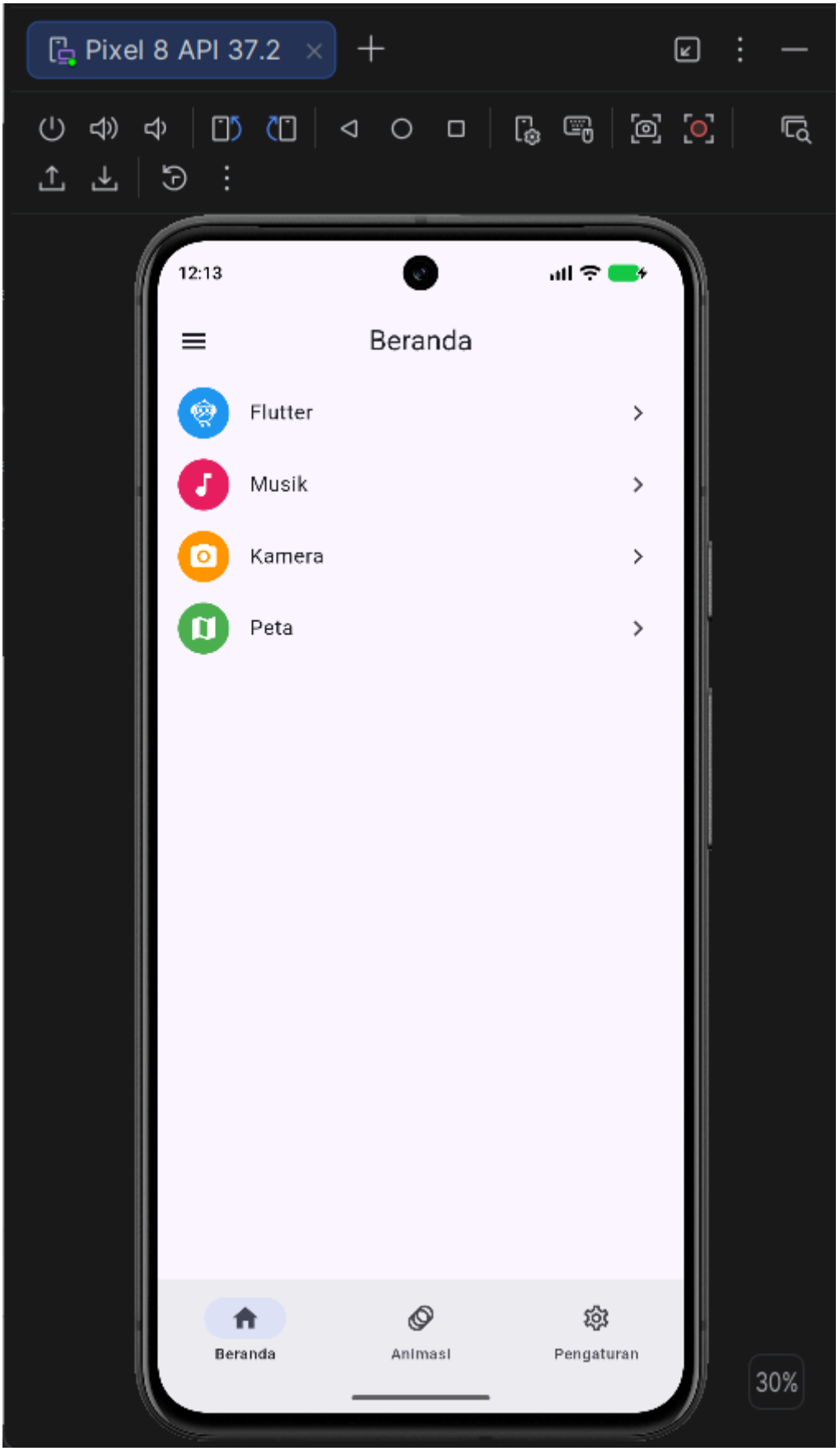
void _pilihTab(int index) {
  setState(() => _index = index);
  Navigator.pop(context);
}

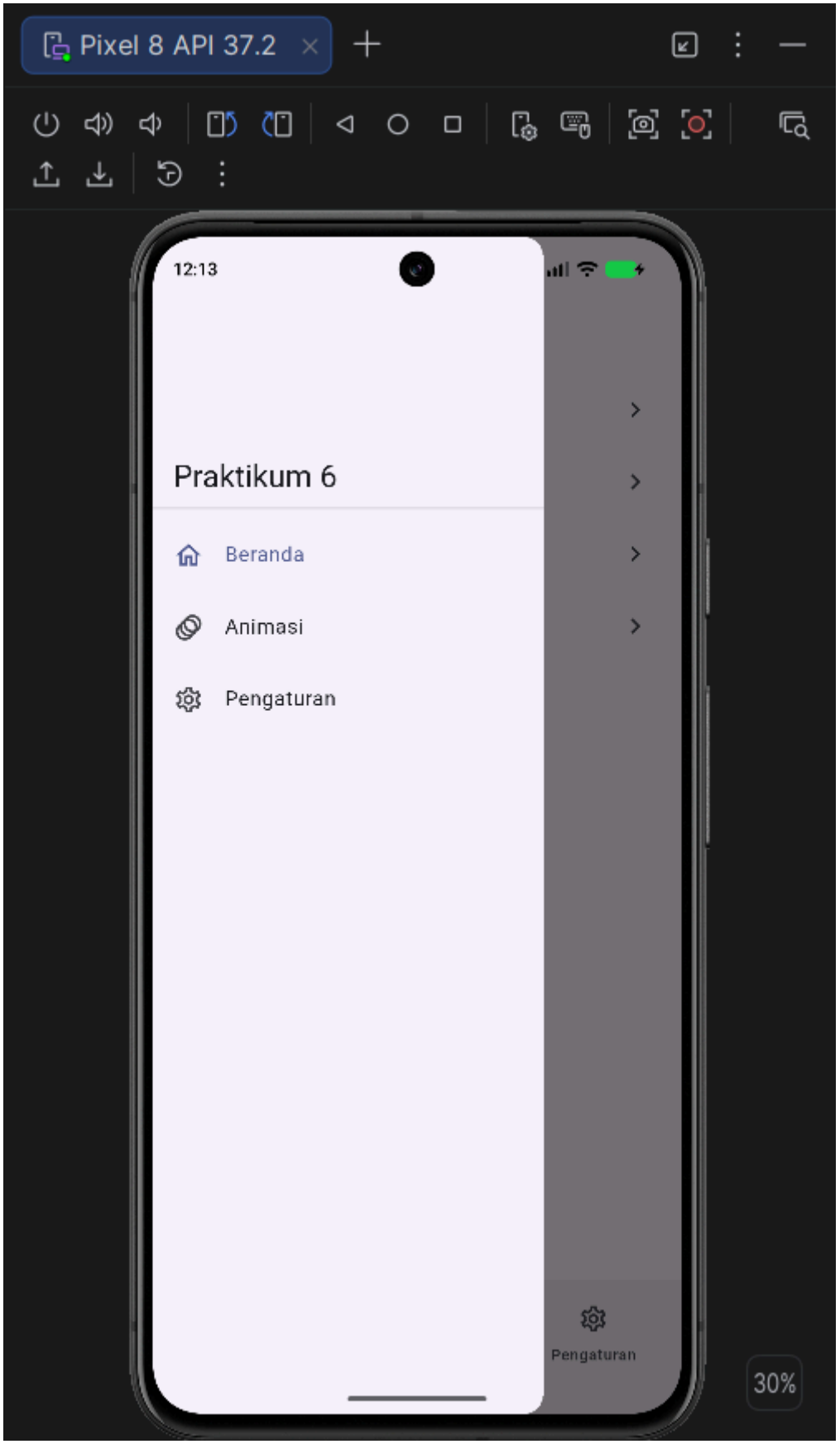
@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(title: Text(_judul[_index])),
    drawer: Drawer(
      child: SafeArea(
        child: ListView(
          children: [
            DrawerHeader(
              child: Align(
                alignment: Alignment.bottomLeft,
                child: Text(
                  'Praktikum 6',
                  style: Theme.of(context).textTheme.headlineSmall,
                ),
              ),
            ),
            ListTile(
              leading: const Icon(Icons.home_outlined),
              title: const Text('Beranda'),
              selected: _index == 0,
              onTap: () => _pilihTab(0),
            ),
            ListTile(
              leading: const Icon(Icons.animation),
              title: const Text('Animasi'),
              selected: _index == 1,
              onTap: () => _pilihTab(1),
            ),
            ListTile(
              leading: const Icon(Icons.settings_outlined),
              title: const Text('Pengaturan'),
              selected: _index == 2,
              onTap: () => _pilihTab(2),
            ),
          ],
        ),
      ),
    ),
    body: IndexedStack(index: _index, children: _halaman),
    bottomNavigationBar: NavigationBar(
```

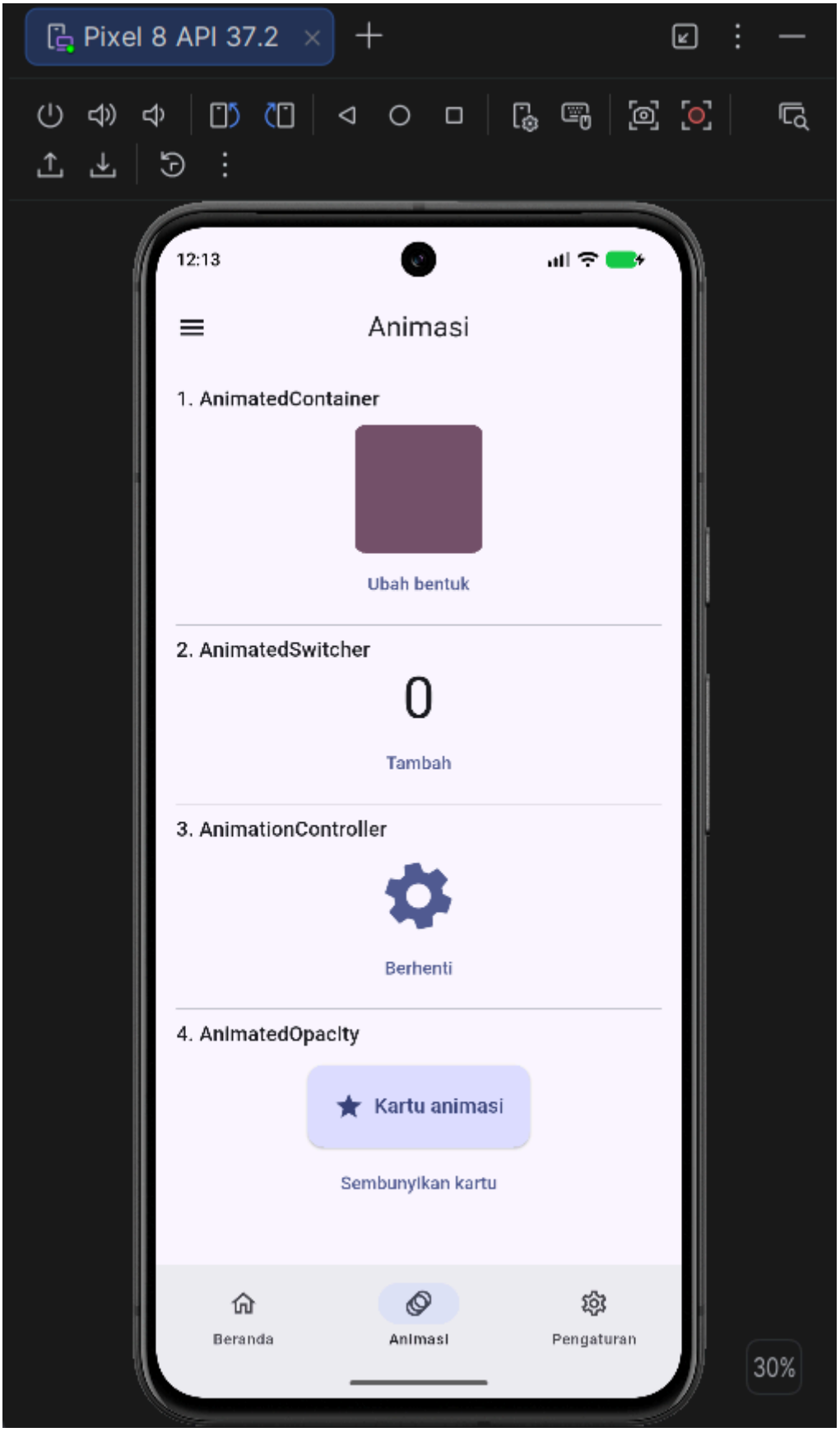
```
      selectedIndex: _index,
      onDestinationSelected: (i) => setState(() => _index = i),
      destinations: const [
        NavigationDestination(
          icon: Icon(Icons.home_outlined),
          selectedIcon: Icon(Icons.home),
          label: 'Beranda',
        ),
        NavigationDestination(
          icon: Icon(Icons.animation),
          label: 'Animasi',
        ),
        NavigationDestination(
          icon: Icon(Icons.settings_outlined),
          selectedIcon: Icon(Icons.settings),
          label: 'Pengaturan',
        ),
      ],
    ),
  );
 }
}

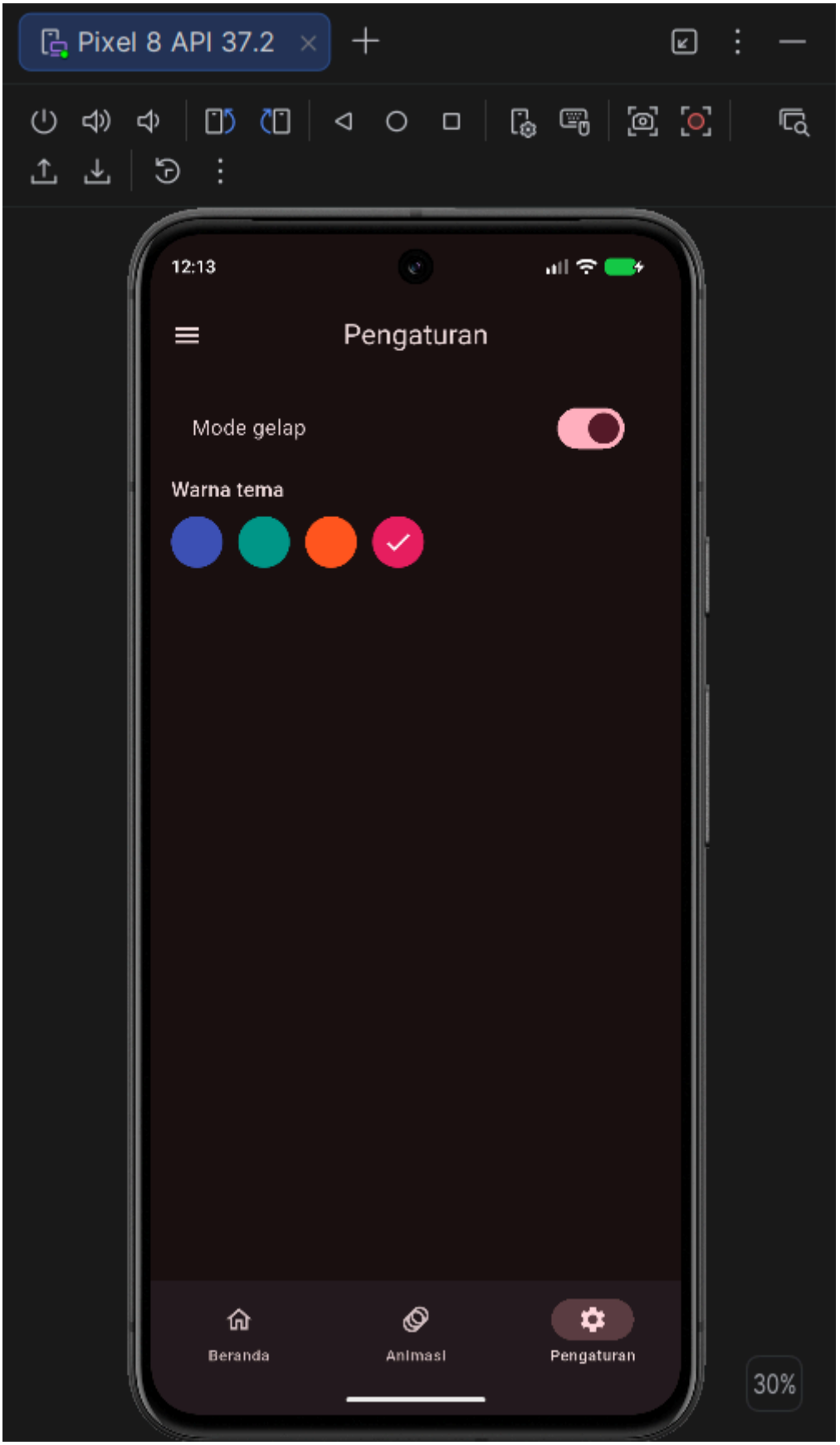
class NotFoundPage extends StatelessWidget {
 const NotFoundPage({super.key});

 @override
 Widget build(BuildContext context) {
   return Scaffold(
     appBar: AppBar(title: const Text('404')),
     body: const Center(
       child: Text('404 - Halaman tidak ditemukan'),
     ),
   );
 }
}
```


Pixel 8 API 37.2
12:13
Beranda
Flutter
Musik
Kamera
Peta
Beranda
Animasi
Pengaturan
30%


Pixel 8 API 37.2
12:13
Praktikum 6
Beranda
Animasi
Pengaturan
Pengaturan
30%


Pixel 8 API 37.2
12:13
Animasi
1. AnimatedContainer
Ubah bentuk
2. AnimatedSwitcher
0
Tambah
3. AnimationController
Berhenti
4. AnimatedOpacity
Kartu animasi
Sembunyikan kartu
Beranda
Animasi
Pengaturan
30%


Pixel 8 API 37.2
12:13
Pengaturan
Mode gelap
Warna tema
Beranda
Animasi
Pengaturan
30%

5. Tugas

Buat aplikasi Galeri/Katalog Interaktif dengan tema bebas (misalnya profil diri, katalog produk, atau galeri tempat wisata) dengan ketentuan:

- Minimal 3 tab memakai NavigationBar (misalnya Beranda, Favorit/Katalog, Pengaturan).
- Named routes dengan argumen untuk halaman detail, dan halaman detail memakai Hero.
- Minimal dua jenis animasi: satu implisit dan satu eksplisit (AnimationController).
- Pengaturan tema terang/gelap dan minimal 3 pilihan warna, tersimpan setelah aplikasi ditutup.
- Warna dan gaya teks mengambil dari Theme.of(context) (hindari warna hard-coded pada teks dan latar).
- Kumpulkan: tangkapan layar (mode terang, mode gelap, halaman detail) dan berkas kode (atau tautan repositori). Aplikasi ini akan menjadi bahan awal untuk mini proyek di Pertemuan 7

```dart
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

final themeMode = ValueNotifier<ThemeMode>(ThemeMode.light);
final seedColor = ValueNotifier<Color>(Colors.indigo);
final daftarFavorit = ValueNotifier<Set<int>>(<int>{});

const List<Color> pilihanWarna = [
 Colors.indigo,
 Colors.teal,
 Colors.deepOrange,
 Colors.pink,
];

class Wisata {
 final int id;
 final String nama;
 final String lokasi;
 final String deskripsi;
 final IconData ikon;

 const Wisata({
   required this.id,
   required this.nama,
   required this.lokasi,
   required this.deskripsi,
   required this.ikon,
 });
}

const daftarWisata = [
 Wisata(
```

```
    id: 1,
    nama: 'Candi Borobudur',
    lokasi: 'Magelang, Jawa Tengah',
    deskripsi:
        'Candi Buddha bersejarah dengan relief dan stupa yang menjadi salah
satu tujuan wisata budaya Indonesia.',
    ikon: Icons.account_balance,
  ),
  Wisata(
    id: 2,
    nama: 'Raja Ampat',
    lokasi: 'Papua Barat Daya',
    deskripsi:
        'Kepulauan dengan perairan jernih, pulau-pulau karst, dan
keanekaragaman kehidupan laut.',
    ikon: Icons.waves,
  ),
  Wisata(
    id: 3,
    nama: 'Gunung Bromo',
    lokasi: 'Jawa Timur',
    deskripsi:
        'Destinasi pegunungan yang dikenal dengan pemandangan kaldera dan
matahari terbit.',
    ikon: Icons.terrain,
  ),
  Wisata(
    id: 4,
    nama: 'Danau Toba',
    lokasi: 'Sumatera Utara',
    deskripsi:
        'Danau vulkanik luas dengan Pulau Samosir dan pemandangan alam di
sekitarnya.',
    ikon: Icons.water,
  ),
];

ThemeData buatTema(Color seed, Brightness brightness) {
  return ThemeData(
    useMaterial3: true,
    colorSchemeSeed: seed,
    brightness: brightness,
    appBarTheme: const AppBarTheme(centerTitle: true),
  );
}

Future<void> main() async {
  WidgetsFlutterBinding.ensureInitialized();

  final prefs = await SharedPreferences.getInstance();
  final modeGelap = prefs.getBool('modeGelap') ?? false;
  final indeksWarna = (prefs.getInt('warnaTema') ?? 0)
      .clamp(0, pilihanWarna.length - 1)
```

```
    .toInt();

 themeMode.value = modeGelap ? ThemeMode.dark : ThemeMode.light;
 seedColor.value = pilihanWarna[indeksWarna];

 runApp(const MyApp());
}

class MyApp extends StatelessWidget {
 const MyApp({super.key});

 @override
 Widget build(BuildContext context) {
   return ListenableBuilder(
     listenable: Listenable.merge([themeMode, seedColor]),
     builder: (context, _) {
       return MaterialApp(
         title: 'Galeri Wisata Indonesia',
         debugShowCheckedModeBanner: false,
         theme: buatTema(seedColor.value, Brightness.light),
         darkTheme: buatTema(seedColor.value, Brightness.dark),
         themeMode: themeMode.value,
         initialRoute: '/',
         routes: {
           '/': (_) => const ShellPage(),
         },
         onGenerateRoute: (settings) {
           if (settings.name == '/detail') {
             final wisata = settings.arguments as Wisata;
             return MaterialPageRoute<void>(
               builder: (_) => DetailPage(wisata: wisata),
               settings: settings,
             );
           }
           return null;
         },
       );
     },
   );
 }
}

void ubahFavorit(Wisata wisata) {
 final favorit = Set<int>.from(daftarFavorit.value);
 if (!favorit.add(wisata.id)) {
   favorit.remove(wisata.id);
 }
 daftarFavorit.value = favorit;
}

class ShellPage extends StatefulWidget {
 const ShellPage({super.key});
```

```dart
 @override
 State<ShellPage> createState() => _ShellPageState();
}

class _ShellPageState extends State<ShellPage> {
 int _index = 0;

 static const _halaman = [
   BerandaTab(),
   FavoritTab(),
   PengaturanTab(),
 ];

 static const _judul = [
   'Beranda',
   'Favorit',
   'Pengaturan',
 ];

 @override
 Widget build(BuildContext context) {
   return Scaffold(
     appBar: AppBar(title: Text(_judul[_index])),
     body: IndexedStack(
       index: _index,
       children: _halaman,
     ),
     bottomNavigationBar: NavigationBar(
       selectedIndex: _index,
       onDestinationSelected: (index) {
         setState(() => _index = index);
       },
       destinations: const [
         NavigationDestination(
           icon: Icon(Icons.home_outlined),
           selectedIcon: Icon(Icons.home),
           label: 'Beranda',
         ),
         NavigationDestination(
           icon: Icon(Icons.favorite_border),
           selectedIcon: Icon(Icons.favorite),
           label: 'Favorit',
         ),
         NavigationDestination(
           icon: Icon(Icons.settings_outlined),
           selectedIcon: Icon(Icons.settings),
           label: 'Pengaturan',
         ),
       ],
     ),
   );
 }
}
```

```dart
class BerandaTab extends StatefulWidget {
 const BerandaTab({super.key});

 @override
 State<BerandaTab> createState() => _BerandaTabState();
}

class _BerandaTabState extends State<BerandaTab>
   with SingleTickerProviderStateMixin {
 bool _besar = false;
 late final AnimationController _putar;

 @override
 void initState() {
   super.initState();
   _putar = AnimationController(
     vsync: this,
     duration: const Duration(seconds: 2),
   )..repeat();
 }

 @override
 void dispose() {
   _putar.dispose();
   super.dispose();
 }

 @override
 Widget build(BuildContext context) {
   final tema = Theme.of(context);
   final warna = tema.colorScheme;

   return ListView(
     padding: const EdgeInsets.all(16),
     children: [
       Text('Jelajahi Indonesia', style: tema.textTheme.headlineSmall),
       const SizedBox(height: 4),
       Text(
         'Temukan destinasi wisata pilihan.',
         style: tema.textTheme.bodyMedium,
       ),
       const SizedBox(height: 16),
       AnimatedContainer(
         duration: const Duration(milliseconds: 400),
         curve: Curves.easeInOut,
         padding: EdgeInsets.all(_besar ? 24 : 16),
         decoration: BoxDecoration(
           color: warna.primaryContainer,
           borderRadius: BorderRadius.circular(_besar ? 28 : 16),
         ),
         child: Row(
           children: [
```

```
            Icon(
              Icons.explore,
              size: _besar ? 42 : 32,
              color: warna.onPrimaryContainer,
            ),
            const SizedBox(width: 12),
            Expanded(
              child: Text(
                'Mulai perjalananmu',
                style: tema.textTheme.titleMedium?.copyWith(
                  color: warna.onPrimaryContainer,
                ),
              ),
            ),
            IconButton(
              onPressed: () => setState(() => _besar = !_besar),
              tooltip: 'Ubah tampilan kartu',
              icon: Icon(
                _besar ? Icons.expand_less : Icons.expand_more,
                color: warna.onPrimaryContainer,
              ),
            ),
          ],
        ),
      ),
      const SizedBox(height: 16),
      Card(
        child: ListTile(
          leading: RotationTransition(
            turns: _putar,
            child: Icon(Icons.explore, color: warna.primary, size: 30),
          ),
          title: const Text('AnimationController'),
          subtitle: const Text('Contoh animasi eksplisit'),
          trailing: IconButton(
            tooltip: _putar.isAnimating ? 'Berhenti' : 'Putar',
            onPressed: () {
              if (_putar.isAnimating) {
                _putar.stop();
              } else {
                _putar.repeat();
              }
              setState(() {});
            },
            icon: Icon(
              _putar.isAnimating ? Icons.pause : Icons.play_arrow,
            ),
          ),
        ),
      ),
      const SizedBox(height: 16),
      Text('Destinasi pilihan', style: tema.textTheme.titleLarge),
      const SizedBox(height: 8),
```

```
      for (final wisata in daftarWisata)
        KartuWisata(wisata: wisata, pakaiHero: true),
    ],
  );
 }
}

class FavoritTab extends StatelessWidget {
 const FavoritTab({super.key});

 @override
 Widget build(BuildContext context) {
  final tema = Theme.of(context);

  return ListenableBuilder(
    listenable: daftarFavorit,
    builder: (context, _) {
      final favorit = daftarWisata
          .where((wisata) => daftarFavorit.value.contains(wisata.id))
          .toList();

      if (favorit.isEmpty) {
        return Center(
          child: Padding(
            padding: const EdgeInsets.all(24),
            child: Text(
              'Belum ada destinasi favorit.\nBuka detail wisata untuk
menambah favorit.',
              textAlign: TextAlign.center,
              style: tema.textTheme.bodyLarge,
            ),
          ),
        );
      }

      return ListView.builder(
        padding: const EdgeInsets.all(16),
        itemCount: favorit.length,
        itemBuilder: (context, index) {
          return KartuWisata(wisata: favorit[index]);
        },
      );
    },
  );
 }
}

class KartuWisata extends StatelessWidget {
 final Wisata wisata;
 final bool pakaiHero;

 const KartuWisata({
  super.key,
```

```
    required this.wisata,
    this.pakaiHero = false,
  });

  @override
  Widget build(BuildContext context) {
    final warna = Theme.of(context).colorScheme;
    Widget ikon = CircleAvatar(
      backgroundColor: warna.secondaryContainer,
      child: Icon(wisata.ikon, color: warna.onSecondaryContainer),
    );

    if (pakaiHero) {
      ikon = Hero(tag: 'wisata-${wisata.id}', child: ikon);
    }

    return Card(
      child: ListTile(
        leading: ikon,
        title: Text(wisata.nama),
        subtitle: Text(wisata.lokasi),
        trailing: const Icon(Icons.chevron_right),
        onTap: () {
          Navigator.pushNamed(
            context,
            '/detail',
            arguments: wisata,
          );
        },
      ),
    );
  }
}

class DetailPage extends StatelessWidget {
  final Wisata wisata;

  const DetailPage({super.key, required this.wisata});

  @override
  Widget build(BuildContext context) {
    final tema = Theme.of(context);
    final warna = tema.colorScheme;

    return Scaffold(
      appBar: AppBar(
        title: Text(wisata.nama),
        actions: [
          ListenableBuilder(
            listenable: daftarFavorit,
            builder: (context, _) {
              final favorit = daftarFavorit.value.contains(wisata.id);
              return IconButton(
```

```
              tooltip: favorit ? 'Hapus dari favorit' : 'Tambah ke
favorit',
              onPressed: () => ubahFavorit(wisata),
              icon: Icon(
                favorit ? Icons.favorite : Icons.favorite_border,
              ),
            );
          },
        ),
      ],
    ),
    body: ListView(
      padding: const EdgeInsets.all(24),
      children: [
        Center(
          child: Hero(
            tag: 'wisata-${wisata.id}',
            child: CircleAvatar(
              radius: 70,
              backgroundColor: warna.primaryContainer,
              child: Icon(
                wisata.ikon,
                size: 68,
                color: warna.onPrimaryContainer,
              ),
            ),
          ),
        ),
        const SizedBox(height: 24),
        Text(
          wisata.nama,
          textAlign: TextAlign.center,
          style: tema.textTheme.headlineMedium,
        ),
        const SizedBox(height: 8),
        Text(
          wisata.lokasi,
          textAlign: TextAlign.center,
          style: tema.textTheme.titleMedium?.copyWith(
            color: warna.primary,
          ),
        ),
        const SizedBox(height: 20),
        Card(
          child: Padding(
            padding: const EdgeInsets.all(16),
            child: Text(
              wisata.deskripsi,
              style: tema.textTheme.bodyLarge,
            ),
          ),
        ),
      ],
```

```
    ),
  );
 }
}

class PengaturanTab extends StatelessWidget {
 const PengaturanTab({super.key});

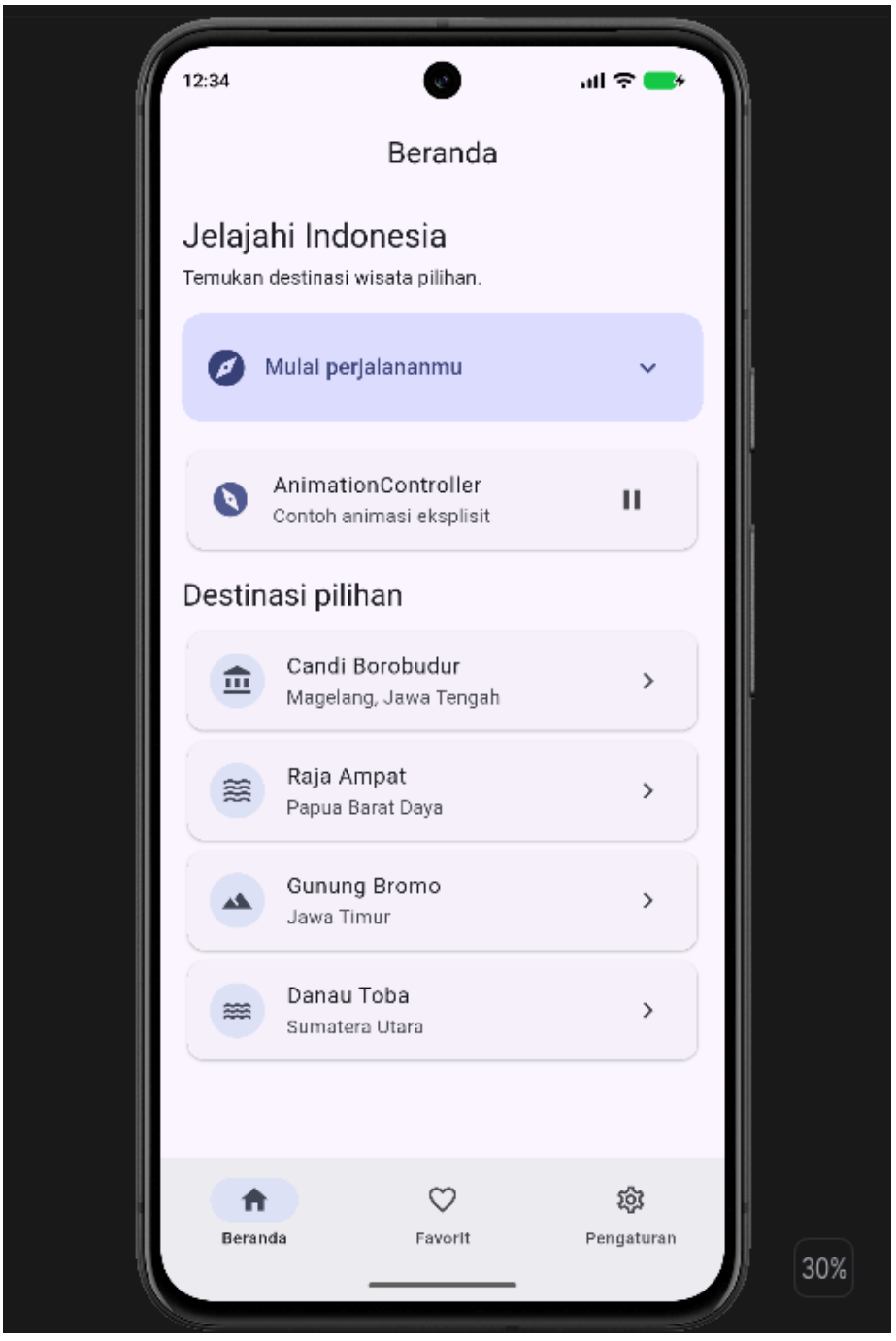
 Future<void> _simpanMode(bool gelap) async {
   final prefs = await SharedPreferences.getInstance();
   await prefs.setBool('modeGelap', gelap);
 }

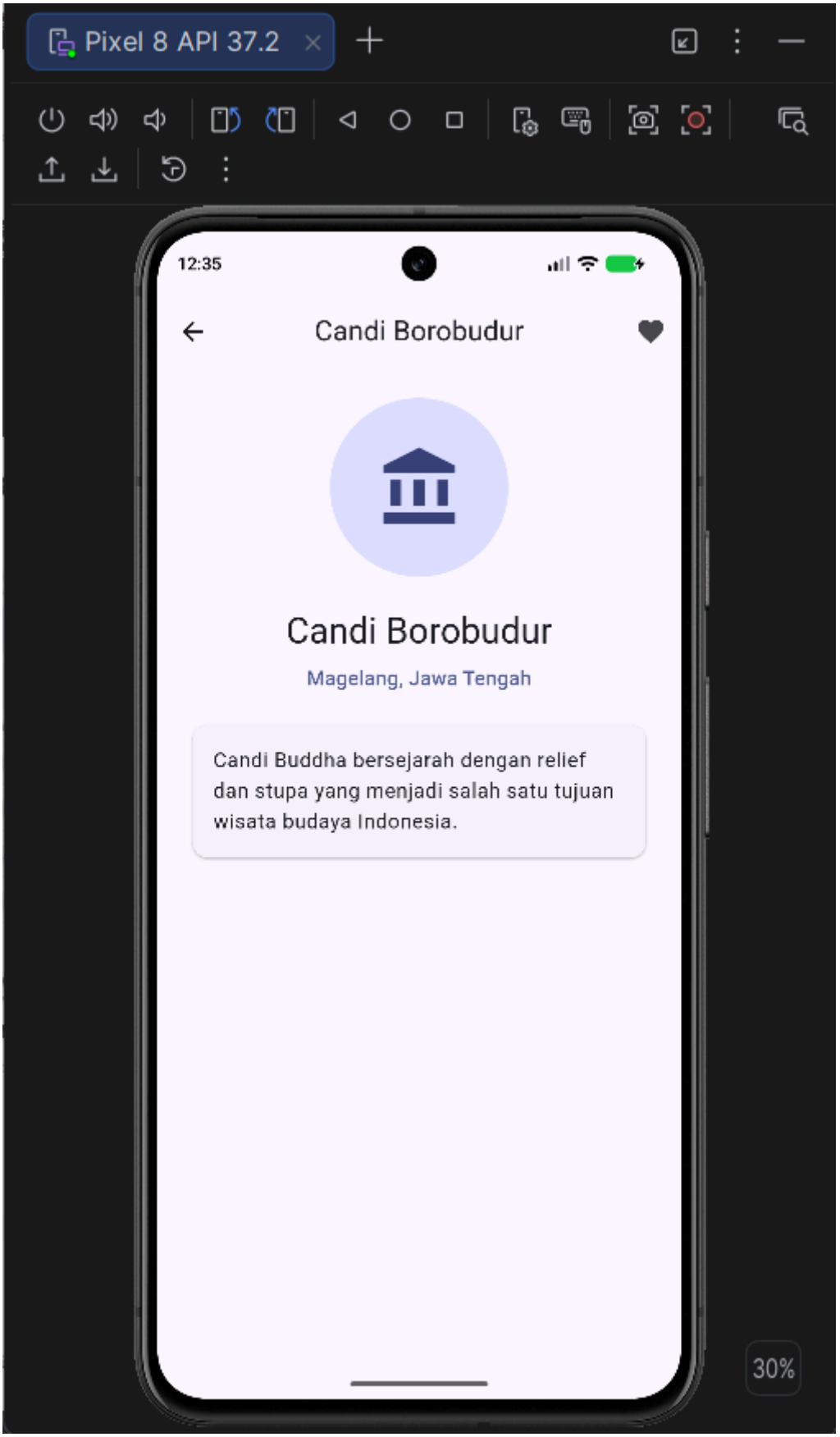
 Future<void> _simpanWarna(Color warna) async {
   final prefs = await SharedPreferences.getInstance();
   await prefs.setInt('warnaTema', pilihanWarna.indexOf(warna));
 }

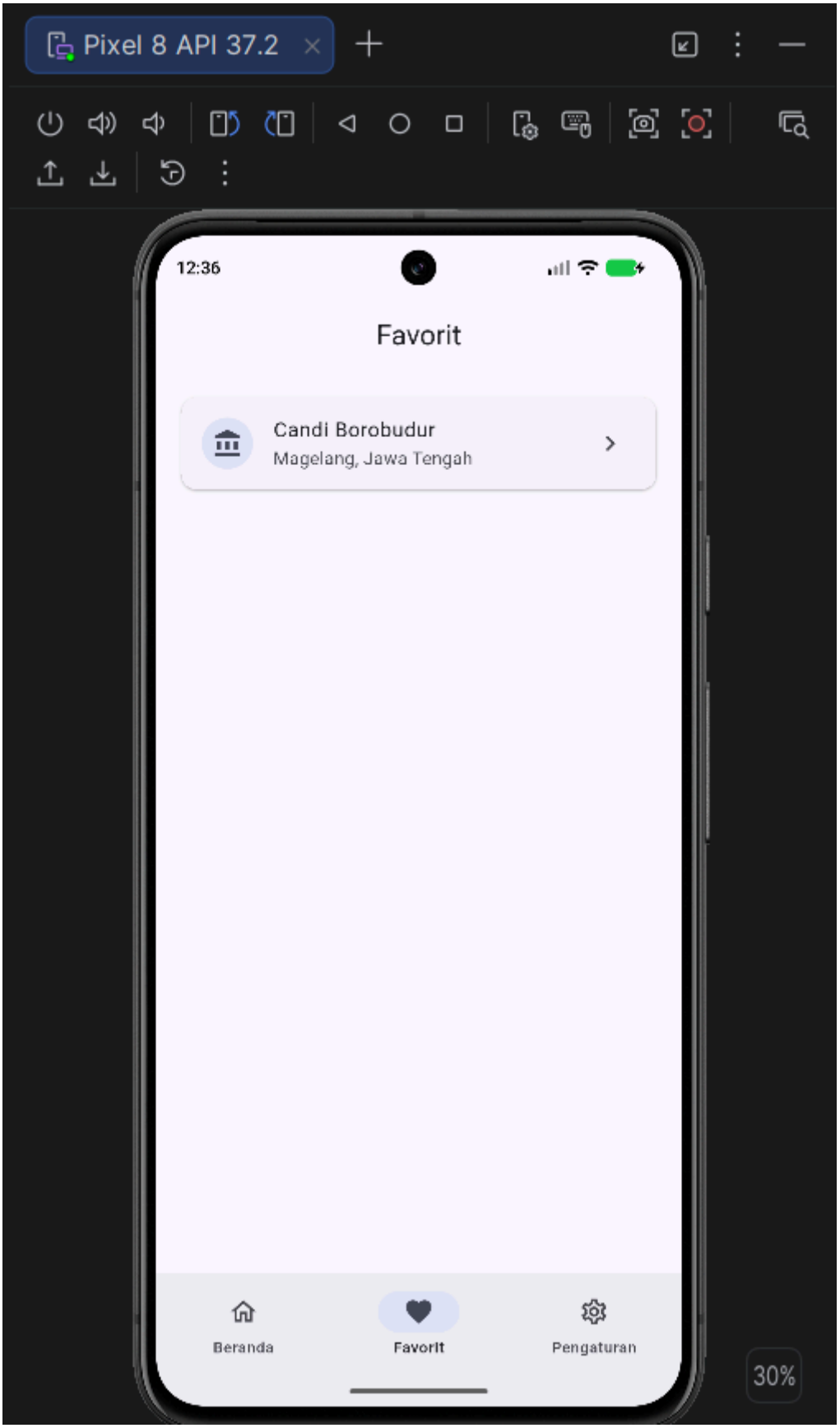
 @override
 Widget build(BuildContext context) {
   final tema = Theme.of(context);
   final warna = tema.colorScheme;

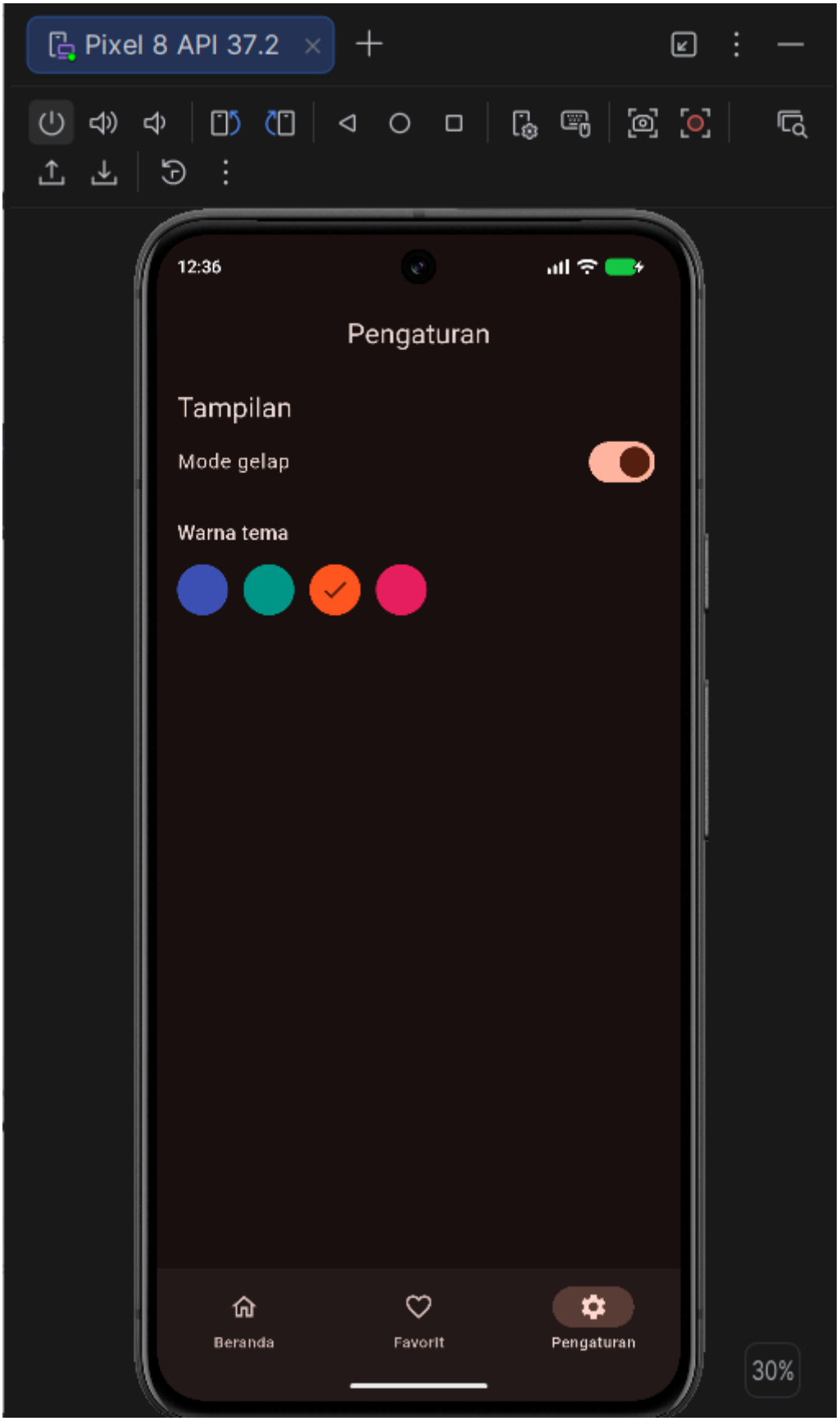
   return ListenableBuilder(
     listenable: Listenable.merge([themeMode, seedColor]),
     builder: (context, _) {
       return ListView(
         padding: const EdgeInsets.all(16),
         children: [
           Text('Tampilan', style: tema.textTheme.titleLarge),
           SwitchListTile(
             contentPadding: EdgeInsets.zero,
             title: const Text('Mode gelap'),
             value: themeMode.value == ThemeMode.dark,
             onChanged: (gelap) {
               themeMode.value = gelap ? ThemeMode.dark : ThemeMode.light;
               _simpanMode(gelap);
             },
           ),
           const SizedBox(height: 16),
           Text('Warna tema', style: tema.textTheme.titleMedium),
           const SizedBox(height: 12),
           Wrap(
             spacing: 12,
             runSpacing: 12,
             children: [
               for (final pilihan in pilihanWarna)
                 InkWell(
                   customBorder: const CircleBorder(),
                   onTap: () {
                     seedColor.value = pilihan;
                     _simpanWarna(pilihan);
                   },
```

```
                    child: CircleAvatar(
                      backgroundColor: pilihan,
                      child: seedColor.value == pilihan
                          ? Icon(Icons.check, color: warna.onPrimary)
                          : null,
                    ),
                  ),
                ],
              ),
            ],
          );
        },
      );
  }
}
```




Pixel 8 API 37.2
12:35
Candi Borobudur
Candi Borobudur
Magelang, Jawa Tengah
Candi Buddha bersejarah dengan relief dan stupa yang menjadi salah satu tujuan wisata budaya Indonesia.
30%


Pixel 8 API 37.2
12:36
Favorit
Candi Borobudur
Magelang, Jawa Tengah
Beranda
Favorit
Pengaturan
30%


Pixel 8 API 37.2
12:36
Pengaturan
Tampilan
Mode gelap
Warna tema
Beranda
Favorit
Pengaturan
30%

6. Pertanyaan Refleksi

1. Apa perbedaan animasi implisit dan eksplisit? Kapan memilih masing-masing?

Animasi implisit seperti AnimatedContainer dan AnimatedSwitcher akan berjalan otomatis ketika nilai atau widget berubah. Animasi eksplisit menggunakan AnimationController sehingga durasi, pengulangan, dan penghentian animasi dapat dikontrol secara langsung. Animasi implisit cocok untuk perubahan tampilan sederhana, sedangkan animasi eksplisit lebih cocok ketika animasi memerlukan kontrol lebih terperinci.

2. Mengapa dua Hero harus memiliki tag yang sama, dan apa yang terjadi bila tag-nya kembali muncul dua kali di halaman yang sama?

Dua widget Hero harus memiliki tag yang sama agar Flutter dapat mencocokkan elemen pada halaman asal dan tujuan untuk membuat transisi. Namun, tag harus unik pada setiap halaman. Jika terdapat dua Hero dengan tag yang sama dalam satu halaman, Flutter dapat mengalami konflik ketika menentukan elemen yang akan dianimasikan.

3. Apa keuntungan memakai Theme.of(context) dibanding menulis warna langsung pada widget?

Theme.of(context) memungkinkan widget menggunakan warna dan gaya yang telah ditentukan dalam tema aplikasi. Dengan begitu, tampilan lebih konsisten dan dapat menyesuaikan diri ketika pengguna mengganti tema terang, gelap, atau warna tema.

4. Apa perbedaan berpindah tab dengan IndexedStack dan membuka halaman dengan Navigator?

IndexedStack digunakan untuk berpindah di antara beberapa tab utama dalam satu halaman sekaligus mempertahankan state masing-masing tab. Sementara itu, Navigator digunakan untuk membuka halaman baru ke dalam tumpukan navigasi dan kembali ke halaman sebelumnya menggunakan pop().

5. Mengapa AnimationController wajib di-dispose?

AnimationController harus di-dispose ketika widget tidak lagi digunakan agar resource animasi dilepaskan dan animasi tidak terus berjalan setelah widget selesai digunakan.

7. Catatan Pribadi

- ThemeData : Digunakan untuk mengatur gaya dan tampilan aplikasi.
- ThemeMode : Menentukan penggunaan tema terang, gelap, atau mengikuti sistem.
- Theme.of(context) : Mengambil warna dan gaya dari tema aplikasi.
- colorSchemeSeed : Menentukan warna dasar untuk menghasilkan palet tema.
- ValueNotifier : Menyimpan nilai yang dapat berubah.
- ListenableBuilder : Membangun ulang widget ketika objek berubah.
- Named Routes : Navigasi halaman menggunakan nama rute.
- Navigator.pushNamed() : Membuka halaman berdasarkan nama rute.
- onGenerateRoute : Menentukan halaman tujuan berdasarkan pengaturan rute.
- Hero : Menganimasikan elemen yang sama ketika berpindah halaman.
- AnimatedContainer : Menganimasikan perubahan ukuran dan warna secara otomatis.
- AnimatedSwitcher : Menganimasikan pergantian widget.
- AnimationController : Mengontrol jalannya animasi secara langsung.
- RotationTransition : Menggunakan animasi untuk memutar widget.
- SingleTickerProviderStateMixin : Menyediakan VSync untuk AnimationController.
- dispose() : Melepaskan resource seperti controller ketika widget tidak lagi digunakan.
- NavigationBar : Membuat navigasi tab di bagian bawah aplikasi.
- NavigationDestination : Mendefinisikan pilihan tab beserta ikon dan labelnya.
- IndexedStack : Menampilkan widget dari beberapa widget dan menahan state tab lain.
- ValueKey : Membantu Flutter mengenali perubahan identitas widget.